

Sources of Labor Market Power*

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Abstract

Wages in the US are typically 30 percent lower than the marginal productivity of labor. A large literature has taken this as suggestive of significant deviations from perfect competition. This paper contrasts two sources of wage markdowns: classical labor market power, arising from imperfect substitution between labor markets, and search labor market power, linked to search frictions that make job finding uncertain. It creates a simple unified theoretical framework and develops empirical tests based on the relationship between posted vacancies and wages, as well as their responses to demand shocks. It finds that search labor market power potentially explains a large share of variation in wage markdowns across firms and primarily drives labor market responses to monetary policy, in contrast to the recent greater emphasis on classical labor market power. These findings suggest it is important to study labor monopsony power in settings with search frictions, where wage markdowns may be partly efficient.

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1 Introduction

Wages in the US are on average 30% lower than the marginal productivity of labor ([Hereshbein et al., 2022](#)). The literature has typically interpreted these as significant deviations from perfect competition, with firms being able to mark down wages. But where do these markdowns come from and are they necessarily inefficient?

The theoretical literature suggests two possible sources of wage markdowns. First, “classical” labor market power, such as in [Berger et al. \(2022\)](#), [Bhaskar et al. \(2002\)](#) and [Robinson \(1969\)](#), stems from imperfect substitution between labor markets. Under this type of monopsony, workers cannot freely transfer from one market to the other due to, for example, specific skill requirements, geographical constraints, and preferences. This in turn enables firms to mark down wages and derive a profit. The second source of wage markdowns is highlighted in the search literature, such as in [Jarosch et al. \(2019\)](#), [Burdett and Mortensen \(1998\)](#), [Moen \(1997\)](#) and [Pissarides \(1985\)](#). In the presence of search frictions, the matching between a worker and a vacancy is not guaranteed. Firms can internalize the value for workers of securing a job and offer lower wages if their vacancies are easier to fill.

Distinguishing these two sources of labor market power has important implications for the transmission of shocks and the efficiency of labor markets, including the desirability of minimum wages. The two sources of labor market power both result in lower wages, but may offer very different predictions for employment levels. While classical labor market power always lowers both wages and employment, search labor market power lowers wages,¹ but has ambiguous employment effects, e.g. no effects (such as in [Jarosch et al. 2019](#) or [Moen 1997](#)), or even larger equilibrium employment (such as in [Rudanko 2023](#)).² Under search labor market power, wage markdowns are thus not sufficient statistics to assess labor market inefficiencies and may sometimes be a fair compensation to employers’ offering of sufficiently high job-finding rate, such as in [Moen \(1997\)](#).

In this paper, we propose a simple framework that allows for classical and search labor market power and develop empirical tests based on the cross-sectional relationship between the number of posted vacancies and wages and the responses of vacancies and wages to demand shocks to ascertain the relative importance of either. The model builds on commonly used “classical” labor market power frameworks by enriching them with a simplified search friction. We first derive model predictions for the case when wage mark-

¹In the paper, we stick with designating the difference between the wage and the marginal productivity of labor due to “search” monopsony as a markdown, even though such a wedge may not be inefficient.

²In standard Diamond-Mortensen-Pissarides models, the relative bargaining power of workers and firms determines a condition for efficiency originally derived by [Hosios \(1990\)](#). Different settings may imply underemployment (when worker bargaining power is too large), overemployment (when firms’ bargaining power is too large) or optimal employment, in which case markdowns fairly compensate employers for raising the likelihood of forming a match.

downs are due to classical monopsony alone, and then we consider how these predictions change when allowing for search monopsony. We test these predictions empirically by studying the relationship between equilibrium wages and vacancies and the response of vacancies and wages to exogenous demand shocks measured by monetary policy surprises.

We find that classical monopsony models cannot rationalize either the relationship between equilibrium vacancies and wages, or the response of labor market variables to monetary policy. In particular, search labor market power dominates classical labor market power by generating a negative relationship between vacancies and wages which would otherwise be positive if classical monopsony was more prevalent. This finding suggests that the variation across firms in equilibrium markdowns is mostly explained by heterogeneity in search rather than classical labor market power. Search labor market power is also crucial for the response of labor market variables to changes in demand, driving the response of vacancies and wages to these shocks.

In the model, each firm has its own labor market, and workers allocate their search across firm-specific labor markets. The substitution between markets is imperfect – leading to the emergence of classical labor market power. Moreover, there are search frictions, making it uncertain for households to land a job. Hence, households take into account both offered wages and the probability of finding a job in each market. Firms can thus offer value to workers in two ways: by raising wages or by offering more certain employment. The ability to generate more certain employment determines what we call search labor power. Classical labor market power is modeled in accordance with standard monopsony frameworks in the recent literature, such as [Berger et al. \(2022\)](#). Search labor market power is introduced in a simplified framework loosely based on the static version of a directed search model by [Moen \(1997\)](#). Unlike the textbook version of directed search models, in our model the substitution between individual labor markets is not perfect, allowing for firms to have both, either, or neither classical and search labor market power.

We first construct a test based on equilibrium wages and vacancies derived from firm decisions. Taking the search decision of workers (i.e. the labor supply) as given, firms choose wages and the number of vacancies to maximize profits. Firms with greater classical market power face an inelastic labor supply because workers cannot easily switch from their labor market. Firms with classical monopsony internalize that raising wages affects their wage bill, and thus post lower wages and fewer vacancies, taking advantage that workers are less sensitive to both vacancies and wages. In contrast, firms with more search market power face workers that are highly reactive to the number of vacancies posted. This makes it relatively easier to hire an additional worker by posting an additional vacancy rather than by raising posted wages. In equilibrium, these firms post relatively more vacancies and pay lower wages. This means that comparing vacancy-level wages and the

relative prevalence of vacancies in the labor market where the vacancy is posted can be a valid test for the most relevant source of monopsony: if wages are positively related to the prevalence of vacancies, that suggests classical monopsony is more important, otherwise search monopsony is more important.

Because equilibrium wages and vacancies may be empirically hard to pin down and suffer from confounding factors, we also derive predictions from responses to exogenous demand shocks. Under classical monopsony power, firms with more market power will not react differentially to demand shocks, unless their wages respond differentially. In contrast, under search labor market power, firms with more market power post relatively more vacancies in response to a demand shock, even if their wages do not adjust differentially. To derive a second test on which source of market power is dominant, we map the predictions of the model to responses that depend on firms' equilibrium wages and vacancies, since we do not observe directly the degree of either classical or search monopsony. The test states that if firms only vary in the extent of classical monopsony alone and wages do not respond differentially, we should expect no heterogeneity in the response of vacancies to monetary policy across firms depending on their equilibrium wages or vacancies. On the contrary, if search labor market power is driving the response, the hiring of firms with lower wages and higher vacancies in equilibrium should respond more to demand shocks. This difference in prediction allows us to test whether classical monopsony is enough to understand the dynamics of labor market variables or whether this response is driven by the search monoposony.

We take these two tests to data on the near universe of online job postings provided by Lightcast (formerly Burning Glass Technologies). Lightcast data cover 250 million online job vacancy postings that correlate strongly with firm-level changes in net employment. Details for each vacancy include information on the firm, location, posted date, job requirements, and offered wage, among others. The highly disaggregated data allow us to construct a firm-local labor market level dataset, effectively allowing for establishment-level data. This dataset is perfectly suited for our purposes, as it focuses on postings by firms as opposed to equilibrium outcomes for employment that are influenced by the search decisions of households. Moreover, the observables in this dataset correspond perfectly to our modeling framework: in the model we assume that search is directed with wages and vacancy postings being observable by the workers, and our dataset exactly contains the information that is available to workers at the time of making search decisions.

We use unexpected high-frequency monetary policy shocks around Federal Open Market Committee (FOMC) meetings from [Jarociński and Karadi \(2020\)](#) to identify exogenous demand shocks. Central banks implement changes in monetary policy, such as adjustments in interest rates or quantitative easing, with the explicit goal of influencing

aggregate demand. Using the unanticipated part of the monetary policy decision allows us to treat them as exogenous variations of aggregate demand. By analyzing vacancies and wages in response to these shocks, we can infer the effects of demand-side disturbances while minimizing concerns about reverse causality or omitted variable biases.

The empirical counter-parts of our model-derived tests offer evidence in favor of the large heterogeneity of search labor market power, and reject important heterogeneity in classical labor market power. We find a negative relationship between equilibrium levels of vacancies and wages and an amplification of the positive effect of accommodative monetary policy on vacancy postings for firms with higher equilibrium vacancies and lower equilibrium wages. Both findings reject the hypothesis that heterogeneity in classical labor market power is a key feature of the data. The effects are significant and large. Quantitatively, a firm at the 50th percentile of vacancy share distribution increases its labor demand by $\approx 7\%$ in response to a 10 basis point surprise monetary loosening while a firm at the 95th percentile of the vacancy share distribution increases labor demand by $\approx 9\%$. Moreover, the effect of monetary policy shocks on firms with market power is persistent, with effects economically large and statistically significant for at least six quarters.

To isolate the effect of labor market power on wages, it is crucial to rule out confounding firm-level differences, such as productivity, size, or regional characteristics. We therefore test whether wages respond differentially to labor market power, both with and without controls. Without controls, we observe significant heterogeneity in wage responses, suggesting that firms with greater labor market power also differ systematically along other dimensions, such as being located in more productive regions or employing different types of workers. However, once we include extensive controls for observable firm and worker characteristics, the heterogeneity in wage responses disappears. This pattern confirms that our main results are not driven by other firm-level differences, but rather reflect the effects of labor market power. In addition, we conduct several other robustness checks, including limiting the sample to only the vacancies that report posted wages, addressing worker heterogeneity by looking at different residualized wage measures, and repeating the results with different monetary policy shocks.

Our paper contributes to several recent branches of the literature. Most notably, our results are important for the studies of labor market power, such as [Berger et al. \(2022\)](#); [Jarosch et al. \(2019\)](#); [Hershbein et al. \(2022\)](#); [Azar et al. \(2019a,b,c, 2020, 2022\)](#); [Ben-melech et al. \(2022\)](#); [Rudanko \(2023\)](#); [Burdett and Mortensen \(1998\)](#); [Pissarides \(1985\)](#). Our paper is the first one to conduct an empirical comparison between the two competing sources of wage markdown and to show that considering only classical labor market power is not enough for a comprehensive understanding of monopsony power in labor markets.

The modeling approach in our paper builds upon the monopsony models of [Berger et al. \(2022\)](#) and directed search, such as in [Moen \(1997\)](#) and classic paper of [Pissarides \(1985\)](#). To our knowledge, our framework is the first to combine both search and classical labor market power in a simple unified framework.

Our approach offers some preliminary insight for the literature on hiring intensity ([Davis et al., 2013](#); [Wolthoff, 2018](#)), aggregate matching elasticity ([Lange and Papageorgiou, 2020](#)), and the Beveridge curve ([Burdett et al., 2001](#)). In particular, cross-firm heterogeneity in the degree of search monopsony would rationalize an amplification of the response of hiring to productivity shocks as in [Wolthoff \(2018\)](#), the findings from [Lange and Papageorgiou \(2020\)](#) that the aggregate matching elasticity is procyclical, and both the flattening and outward shift of Beveridge curve, documented by [Burdett et al. \(2001\)](#).

Our paper also contributes to the literature on the effects of monetary policy on labor markets, in particular [Romer and Romer \(1989\)](#); [Fornaro and Wolf \(2021\)](#); [Coglianese et al. \(2021\)](#); [Dolado et al. \(2021\)](#); [Coibion et al. \(2017\)](#); [Andersen et al. \(2021\)](#); [Jasova et al. \(2021\)](#); [Bartscher et al. \(2021\)](#); [Bergman et al. \(2022\)](#). It offers novel results on the response of vacancies and wages to monetary policy, and finds that effects coming from search labor market power are quantitatively significant, persistent, and are different from the ones suggested by the common intuition based on classical labor market power. These effects are of critical interest for monetary policy makers that need to target labor market outcomes, such as unemployment, market tightness, or wage growth.

Finally, our paper relates to the literature that uses data from Lightcast, formerly known as Burning Glass Technologies (BGT). Lightcast is among the best established datasets for vacancy postings. Papers that specifically looked at labor market power using this dataset include [Hershbein and Kahn \(2018\)](#); [Hazell et al. \(2021\)](#); [Hershbein et al. \(2022\)](#); [Azar et al. \(2022\)](#). Those papers mostly focus on the equilibrium effects of labor market power, such as the levels of wages, and do not explore either the specific frictions behind the markdowns or the effects of monetary policy on labor markets.

The remainder of this paper is organized as follows: Section 2 lays out a search and matching model and formulates tests on the relative prevalence of search and classical monopsony. Section 3 introduces our data and Section 4 details our empirical approach and results. Finally, Section 5 concludes.

2 Theory

This section introduces a model with both classical and search labor market power. To model classical labor market power, we follow the traditional macro approach in the vein of [Berger et al. \(2022\)](#) and combine it with simple elements of a directed search framework

as in [Moen \(1997\)](#). We then derive predictions for equilibrium vacancies and wages and their responses to demand shocks, which we test in [Section 4](#).

General Environment The economy consists of a representative household and a unit mass of firms that are heterogeneous along three dimensions: 1) the degree of substitution across labor markets (governing classical monopsony), 2) firms' marginal ability to hire when posting vacancies (governing search monopsony), and 3) their marginal revenue of labor, which summarizes many other important firm characteristics, such as productivity, capital depth, and markups.

Firms hire workers in firm-specific labor markets and produce a differentiable and imperfectly substitutable good to maximize instantaneous profit. Labor markets are subject to search frictions: firms post vacancies and wages and are uncertain about how many workers they will be able to hire. As in the tradition of macroeconomic models (but not usually in directed search models), labor is assumed to be a flexible input, so that firms need to re-hire every period. This assumption greatly aids in considering the two types of monopsony power in a single, tractable framework. Workers decide on their consumption bundle, choose search effort, and are similarly uncertain about whether they can find employment.

Two Sources of Labor Monopsony When choosing their search effort, workers see local labor markets as substitutable based on parameter γ_i , where i denotes firm-specific labor markets. γ_i is the first source of labor market power: if workers see a firm's labor market as less substitutable, that firm can take advantage of this and mark down wages. This reflects *classical* labor market power based on consumer preferences and imperfect substitution between labor markets (see, for instance, [Berger et al. 2022](#) or [Card et al. 2018](#)).³

In addition, in the presence of search frictions not all workers who search for a job at firm i will be hired. Assuming a Cobb-Douglas matching function, firm i 's total hiring is given by $l_{it} = \left(\frac{v_{it}}{n'_{it}}\right)^{\alpha_i} n'_{it}$, where n'_{it} is the total search time spent by all workers directed at firm i and v_{it} are the vacancies posted by firm i , and $\left(\frac{v_{it}}{n'_{it}}\right)^{\alpha_i}$ is the matching rate. α_i is the second source of labor market power, determining how much more likely a firm is to hire a worker if it posts one additional vacancy. A high search labor market power firm, or one with high α_i , is able to offer welfare to the worker through a higher matching rate, which then allows the firm to pay lower wages as shown below. We label this second source of labor market power as *search* labor market power.

³In the baseline, we assume that elasticities are constant over time and, in particular, do not depend on search efforts directed at each firm. In an extension, we relax this assumption.

Search labor market power can be microfounded as deriving from hiring intensity, which has been found to be a crucial factor driving employment growth at the establishment level (Davis et al., 2013). Firms can adjust hiring intensity by increasing advertising, screening candidates more efficiently, or leveraging economies of scale. The first two align with our model of a fixed parameter α_i that varies across firms, while the latter suggests α_i increases with vacancies, which would result in similar predictions as in our baseline model. Wolthoff (2018) offers a microfounded model of this mechanism, which leads to an amplification of the hiring response just as predicted by our much simpler framework. Lange and Papageorgiou (2020) additionally predicts that the aggregate matching elasticity, α , is procyclical, which is consistent with the hiring of higher-alpha firms being more reactive to demand shocks, which would exactly lead to procyclical average alpha due to a compositional effect. An alternative explanation of search labor market power can be due to differing workers' tolerance for uncertain job-finding probabilities. A firm that faces a pool of applicants that values higher job-finding rates can offer lower wages by posting more vacancies (Barnichon and Figura, 2015; Abbritti and Consolo, 2024).

Household Problem We assume the representative household maximizes a utility function that is separable in labor and consumption, which allows us henceforth to focus on labor market decisions exclusively. The household chooses how much to search (n_{it}), taking as given wages (w_{it}), total search to firm i (n'_{it}), aggregate search across all labor markets (N_t), and vacancies posted by firm i (v_{it}) according to the following maximization problem⁴:

$$W_t L_t = \max_{n_{it}} \int \left(\frac{v_{it}}{n'_{it}} \right)^{\alpha_i} n_{it} w_{it} di$$

$$\int \left(\frac{n_{it}}{N_t} \right)^{1+\gamma_i} di = 1$$

where we replaced labor at any given firm with the hiring function introduced earlier comprising the matching rate $\left(\frac{v_{it}}{n'_{it}} \right)^{\alpha_i}$, which is taken as given by the consumer who does not internalize the “crowding-out” effect of her search efforts, multiplied by the amount of search she chooses.

The following first-order condition determines optimal search in each firm's labor market:

$$\left(\frac{v_{it}}{n'_{it}} \right)^{\alpha_i} \frac{w_{it}}{W_t} = L_t (1 + \gamma_i) \left(\frac{n_{it}}{N_t} \right)^{\gamma_i} \frac{D_t}{N_t}$$

⁴With slight abuse of notation, we write down all the integrals with respect to the firm type i without specifying the distribution firm types i and integrating with respect to that measure.

where $D_t^{-1} = \int_0^1 (1 + \gamma_i) \left(\frac{n_{it}}{N_t} \right)^{1+\gamma_i} di$, is a dispersion term. For the above first-order condition, we have used that $n'_{it} = n_{it}$ in equilibrium, because our household is representative. This optimal search condition defines labor supply in this model. Note that labor supply adjusts to two margins. First, search depends positively on wages, i.e. higher wages in firm i motivate workers to supply more labor to that firm. This is the only margin of adjustment in macroeconomic models without search frictions. Second, due to search frictions, finding a job is uncertain, and thus even if firm i offers high wages, the worker may not direct her search there if the probability of securing a job is too low. Therefore, potential workers trade off the likelihood of finding a job against offered wages. A firm can attract more workers by posting more vacancies, because this raises the probability that any worker will find a job, which is valued by workers.

Firms' Problem Firms maximize profits by choosing which wage to offer and how many vacancies to post, taking the demand for their goods and the amount of search allocated to their labor market as given. Their revenue is given by $p_{it}y_{it}$ and depends on the demand for the firm's goods and production function. For this partial equilibrium set-up, we abstract from specifying the demand function and represent it by $y_{it} = \mathcal{Y}_{it}$, where $\mathcal{Y}_{it}$ depends on the firm's relative price, aggregate production and, possibly, other parameters. Aggregate production itself is a function $f(l, I)$ of labor and possibly other inputs. The firm's costs include payroll $w_{it}l_{it}$, vacancy-posting costs cv_{it} ,⁵ and expenditures unrelated to labor Q_{it} .

The profit maximization problem for firm i is:

$$\begin{aligned} \max_{v_{it}, w_{it}} & p_{it}y_{it} - w_{it}l_{it} - cv_{it} - Q_{it} \\ l_{it} &= \left(\frac{v_{it}}{n_{it}} \right)^{\alpha_i} n_{it} \\ \left(\frac{v_{it}}{n_{it}} \right)^{\alpha_i} \frac{w_{it}}{W_t} &= L_t(1 + \gamma_i) \left(\frac{n_{it}}{N_t} \right)^{\gamma_i} \frac{D_t}{N_t} \\ y_{it} &= \mathcal{Y}_{it} \\ y_{it} &= f(l_{it}, I_{it}) \end{aligned}$$

The first constraint denotes the hiring function introduced earlier; the second constraint denotes the optimal labor supply decision from the household problem; and the third and fourth constraints denote the demand and the production function, respectively. Note that this problem allows for capital to be an additional input into production (through

⁵The cost of vacancies can be thought as the cost of sifting through applicants' CVs and doing interviews, as well as expenses tied with incorporating a worker in the firm more generally like training. This cost should not be interpreted literally, as posting a vacancy should be relatively cheap.

the parameter vector I_{it} and investment costs are included in Q_{it}). The only assumption made is that labor and capital spending are separable in the cost function, so that Q_{it} does not depend on labor.

The first-order conditions of the problem above are:

$$\begin{aligned} v_{it} &= \frac{l_{it}}{c} \frac{\xi_i^v}{\xi_i^w + 1} m_{it} \\ w_{it} &= \frac{\xi_i^w}{\xi_i^w + 1} m_{it} \end{aligned}$$

where $\xi^w = \frac{\partial l_{it}}{\partial w_{it}} \frac{w_{it}}{l_{it}}$ is the elasticity of labor supply with respect to wage, $\xi^v = \frac{\partial l_{it}}{\partial v_{it}} \frac{v_{it}}{l_{it}}$ is the elasticity of labor supply with respect to vacancies, and $m_{it} = (p_{it} y_{it})'_l$ is the marginal revenue of labor. Note that the marginal revenue of labor depends on labor productivity (which is a function of I_{it}) and on the firm's markup in the product market (which would be reflected in the product demand $\mathcal{Y}_{it}$). For labor market outcomes, it does not matter whether higher marginal revenue of labor comes from higher productivity or higher product market power, because these only influence the hiring decision through m_{it} .

Solving for the elasticities using the household optimal search equation, the first order conditions above simplify to:

$$v_{it} = \alpha_i \frac{l_{it}}{c} m_{it} \tag{1}$$

$$w_{it} = \frac{1 - \alpha_i}{\gamma_i + 1} m_{it} \tag{2}$$

where the elasticities of labor supply with respect to vacancies and wages are, respectively: $\xi_i^v = \alpha_i \frac{\gamma_i + 1}{\alpha_i + \gamma_i}$ and $\xi_i^w = \frac{1 - \alpha_i}{\alpha_i + \gamma_i}$. The latter is the same as in equation (9) of [Manning \(2021\)](#) using a different framework based on directed search and the amenities model of [Card et al. \(2018\)](#).

The markdown, or the extent to which firms pay wages below the marginal revenue of labor, is given by a modified Lerner formula ([Lerner, 1934](#)):

$$\nu_i = \frac{\xi_i^w}{\xi_i^w + 1} = \frac{1 - \alpha_i}{\gamma_i + 1} \tag{3}$$

The two sources of monopsony introduced earlier are clearly visible in the formula for the markdown. The first derives from *classical* monopsony and is more commonly discussed in the literature (e.g., [Berger et al. 2022](#)). It is represented by γ_i , which determines the inverse of the elasticity of substitution between labor markets for the worker.⁶ Under classical monopsony, wages depend negatively on γ_i , meaning that firms are able to pay lower wages if switching between labor markets is difficult. The second source

⁶Under $\alpha_i = 0$, one obtains the familiar markdown formula from the classical monopsony literature.

of monopsony power is due to the presence of search frictions and is reflected in the parameter α_i . Under search monopsony, wages depend negatively on the rate through which a firm converts vacancies to hiring as governed by α_i . Firms are able to pay lower wages because they can offer an alternative source of value to workers: by posting vacancies, they can reduce the likelihood of unemployment for workers searching in their labor market. Markdowns arise from a combination of these two forces. The following two propositions connect these two sources of labor market power to vacancy postings. These are formulated using calibrated parameters from the literature.

Proposition 1 (classical monopsony). *Firms with greater classical monopsony power (larger γ_i) post **less** vacancies and pay lower wages in equilibrium.*

Proof. See Appendix B.1. □

Proposition 2 (search monopsony). *Firms with greater search monopsony (larger α_i) pay lower wages, but post **more** vacancies in equilibrium.*

Proof. See Appendix B.2. □

Propositions 1 and 2 show that both sources of labor market power allow firms to mark down wages, as also evident from equation (3), but have opposite effects on vacancies. Firms with classical monopsony internalize that workers cannot easily switch away from their labor market and take advantage of that by offering lower wages and lower vacancies, which are both costly for the firm and beneficial for the worker. Firms with search monopsony offer more vacancies to the extent that is cheaper than raising wages.

Propositions 1 and 2 suggest the following preliminary test of the relative prevalence of the two sources of labor market power:

Test 1: In equilibrium, classical monopsony implies a positive relationship between vacancies and wages, while search monopsony implies a negative relationship. This is tested in Section 4.1.

2.1 Monetary Policy Response

Empirically, it is difficult to disentangle the sources of labor market power in equilibrium. In this subsection, we show how the relative importance of classical and search labor market power can be assessed using the response of vacancy postings and wages to exogenous demand shocks. To do so, we use exogenous monetary policy shocks.

Monetary policy shocks are widely regarded as effective instruments for identifying exogenous demand shocks. Central banks implement changes in monetary policy, such as

adjustments in interest rates or quantitative easing, with the explicit goal of influencing aggregate demand. Using the unanticipated part of the monetary policy decision allows us to treat them as exogenous variations of aggregate demand. By analyzing vacancies and wages in response to these shocks, we can infer the effects of demand-side disturbances while minimizing concerns about reverse causality or omitted variable biases.

In this environment, monetary policy does not affect hiring directly. Instead, monetary policy affects the hiring problem indirectly through the marginal revenue of labor, or m_{it} . This follows from assuming a static hiring problem and means that monetary policy only affects household consumption (the demand for firm's goods) and firms' investment.

Log-linearizing around the equilibrium in equations (1)-(2) and considering a small change in m_i denoted with a hat, given the response of wages $\hat{w}_i$, the contemporaneous responses of vacancies is given by:

$$\hat{v}_i = \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i} di} \right) + \hat{N} \quad (4)$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$.

Note that the equations above allow for three types of heterogeneity: the two sources of labor market power, γ_i and α_i , and factors not related to labor market power, reflected in m_i .

Proposition 3 (monetary policy response). *In a model without search monopsony and following a monetary policy shock, if firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, if firms have varying degrees of search monopsony power and do not change their wages differentially, vacancies of firms with greater search monopsony power react more.*

Proof. See Appendix B.3. □

Crucially for Proposition 3 is that wages are not responding differentially to either classical or search monopsony. Under this condition, two firms with the same level of search labor market power and different levels of classical labor market power would also not post vacancies differentially because the elasticity of labor supply to vacancies is the same for both. On the contrary, if there are two firms that differ in search monoposony power, they would adjust their vacancies differentially, because the firm with greater search labor market power can gain more by posting one additional vacancy.

We would also note that if wages respond homogeneously across firms with varying degrees of either classical or search monopsony power, that rules out that differences in the

marginal revenue of labor would be the driving factor behind the response of vacancies, because any such differences would be reflected in the response of wages.

Appendix B.4 shows that Proposition 3 is robust to allowing γ_i to depend on the own search share $\frac{n_i}{N}$, under the assumption that the substitution between markets is homogeneous of degree one. This expands the baseline model to include Kimball and Atkeson-Burstein substitution between labor markets, with the latter including, among others, the model in Berger et al. (2022). Moreover, Appendix B.5 shows that Proposition 3 is also robust to non-constant returns to scale or imperfect competition in the product market.

Importantly, Appendix B.6 shows that Proposition 3 is robust to assuming wages are rigid as long as wages respond homogeneously to varying degrees of both classical and search monopsony power.⁷ This is relevant since even though there is some evidence of relative flexibility of wages of new hires.⁸

To test Proposition 3, we formulate a test based on responses to monetary policy shocks using *equilibrium vacancies and wages*. While we do not observe directly the degree of either search or classical labor market power in the data and estimates of either parameter are scant in the literature, Proposition 1 and Proposition 2 directly maps equilibrium vacancies and wages to the degree of both classical and search labor market power.

In what follows, we map equation (4) to be a function of equilibrium vacancies and wages rather than directly depend on α_i and γ_i :

$$\hat{v}_i = \beta^{vv} \text{ equilibrium vacancies} \times \hat{m}_i \quad (5)$$

$$\hat{v}_i = \beta^{vw} \text{ equilibrium wages} \times \hat{m}_i \quad (6)$$

Matching equation (4) with (5) and taking a first order approximation of the interaction term while appealing to the implicit function theorem, β^{vv} can be decomposed into the following two terms:

$$\beta^{vv} = \underbrace{\frac{1}{(1-\alpha_i)^2} \left(1 + \frac{1}{\gamma_i}\right) \alpha'_v}_{\text{effect of } \alpha} - \underbrace{\frac{1}{\gamma_i^2} \left(\frac{\alpha_i}{1-\alpha_i} - \frac{\int \lambda_i \frac{1}{\gamma_i} \frac{\alpha_i}{1-\alpha_i} di}{\int \lambda_i \frac{1}{\gamma_i} di} \right) \gamma'_v}_{\text{effect of } \gamma}$$

where $\alpha'_v = \frac{-w'_\gamma}{v'_\gamma w'_\alpha - v'_\alpha w'_\gamma}$ and $\gamma'_v = \frac{w'_\alpha}{v'_\gamma w'_\alpha - v'_\alpha w'_\gamma}$ are partial derivatives obtained by considering

⁷Note that no additional assumptions on the wage setting mechanism are needed, so the proposition holds under time-dependent (*a la* Calvo) and state-dependent (menu costs) frameworks.

⁸See Haefke and Sonntag (2013) and Martins et al. (2012). More recent papers, such as Grigsby et al. (2021) and Hazell and Taska (2020), have found evidence of substantial wage rigidities even for new hires. Grigsby et al. (2021) shows that time-dependence (similar to Calvo pricing) is a good approximation of how wages adjust outside recessions.

a reverse mapping from equilibrium wages and vacancies to parameters α, γ , while keeping m constant. Per Propositions 1 and 2, $\alpha'_v > 0$ and $\gamma'_v < 0$. Thus, the effect of alpha on β^{vv} is always positive, while the effect of γ is ambiguous. If the firm has very little search market power compared to other firms, β^{vv} could be negative, while if the firm has significant relative search market power the total effect would be positive.

Matching equation (4) with (6) and taking a first order approximation of the interaction term while appealing to the implicit function theorem, β^{vw} can be decomposed into the following two terms:

$$\beta^{vw} = \underbrace{\frac{1}{(1 - \alpha_i)^2} \left(1 + \frac{1}{\gamma_i}\right) \alpha'_w}_{\text{effect of } \alpha} - \underbrace{\frac{1}{\gamma_i^2} \left(\frac{\alpha_i}{1 - \alpha_i} - \frac{\int \lambda_i \frac{1}{\gamma_i} \frac{\alpha_i}{1 - \alpha_i} di}{\int \lambda_i \frac{1}{\gamma_i} di}\right) \gamma'_w}_{\text{effect of } \gamma}$$

Where $\alpha'_w = \frac{v'_\gamma}{v'_\gamma w'_\alpha - v'_\alpha w'_\gamma}$ and $\gamma'_w = \frac{-v'_\alpha}{v'_\gamma w'_\alpha - v'_\alpha w'_\gamma}$ are again partial derivatives obtained by considering a reverse mapping from equilibrium wages and vacancies to parameters α, γ , while keeping the m constant. Per Propositions 1 and 2, $\alpha'_w < 0$ and $\gamma'_w < 0$. Thus, the effect of alpha on β^{vw} is always negative, while the effect of γ is again ambiguous. If the firm has very little search market power compared to other firms, β^{vw} could be negative, while if the firm has significant relative search market power the total effect could turn positive.

With these effects in mind, we are now ready to formulate our second test:

Test 2: If wages of all firms respond homogeneously to monetary policy and vacancies respond more in firms with higher equilibrium vacancies and lower equilibrium wages, the model where firms vary only in their extent of classical monopsony can be rejected. This implies that search monopsony drives the response of wages and vacancies to monetary policy. This is tested in Section 4.2.

3 Data

3.1 Lightcast data

Lightcast data, formerly Burning Glass Technologies, tracks online vacancy postings from over 45,000 online job boards, carefully removes duplicates, and cleans the data. The resulting dataset covers the near universe ($\approx 70\%$) of all U.S. vacancy postings and comprises ≈ 250 million job vacancy postings for the years of 2007 and 2010-2019.

Lightcast has an extensive coverage across differently sized firms. Unlike survey data, it is collected directly from firms' postings and therefore is a more accurate representation

of the vacancies in the economy. Concretely, it is free from the limitations of datasets that only cover firms of a certain size or firms that satisfy certain criteria, such as being publicly traded like Compustat.

Lightcast can be used as an establishment level data due to extensive geographical breakdown of posted vacancies. Each vacancy contains a FIPS county code, which allows it to be mapped onto any other geographical unities, such as states, MSA or, in our case Commuting Zones.

Lightcast data also offers other significant details on the type of vacancy. NAICS industries, and ONET occupation breakdowns are available. A large proportion of vacancies also lists job requirements, such as education or software skills. Education is reported for approximately half of vacancies. When education is missing, we impute it based on the data for the existing vacancies using the finest occupational breakdown. Effectively we assign the same education requirement within the same occupation. This procedure eliminates most of the missing values.

Lightcast vacancy data has some shortcomings due to the way it is collected, especially in earlier years. The main concern is that online vacancy postings are not representative of all the postings in the economy with an over-representation of certain industries, such as IT or Education. However, robustness checks, for instance in [Hershbein and Kahn \(2018\)](#) indicate that, despite these shortcomings, the resulting vacancy data tracks aggregate and industry trends closely.

Lightcast wage data is significantly less extensive than vacancy data, with only 17% of the vacancies reporting wages. [Hazell et al. \(2021\)](#) find that this limitation does not preclude the data from being representative, with median posted wages from Lightcast closely tracking the corresponding statistics in Occupational Employment Statistics (OES). Some postings list a wage range — in these instances, we take the midpoint of that range. For most of our analysis, we collapse vacancy-level data into a panel of firm-, commuting zone- and quarter-level, or effectively an establishment-level panel. Wage data on annual compensation and does not include bonuses and other benefits beyond the basic wage.

Definition of local labor market. Our analysis mainly compares responses of establishments within the same labor market, so it is crucial to both define a local labor market and come up with equilibrium vacancy and wage measures that are consistent. We define a local labor market as a commuting zone, because it is a closer representation of the options available to a worker searching for job in a particular labor market.

Measure of equilibrium vacancies and wages. As discussed below, we use cumulative vacancy shares as an equilibrium measure of vacancies and cumulative wage residual,

cleaned from occupational effects, as an equilibrium measure of wages.

The measure of the equilibrium level of vacancies is the cumulative share of vacancies posted by a single firm in a local labor market out of the total vacancies posted in that labor market:

$$\text{Equilibrium Vacancy Share}_{i,c,t} = \frac{\sum_{\tau < t} v_{i,c,\tau}}{\sum_{\tau < t} \sum_i v_{i,c,\tau}} \quad (7)$$

where vacancies in commuting zone c , for firm i at time τ are denoted by $v_{i,c,\tau}$.

To measure the equilibrium level of wages we need to account for the significant heterogeneity across occupations and geography. First, we residualize wages by running a fixed-effect regression and canceling out any occupation and commuting zone related factors; and aggregate the results to firm-CZ-time level. We then compute a vacancy-weighted cumulative wage measure using the formula below and normalize the result:

$$\text{Equilibrium Wage Measure}_{i,c,t} = \frac{\sum_{\tau < t} v_{i,c,\tau} \times \text{residual wages}_{i,c,\tau}}{\sum_{\tau < t} v_{i,c,\tau}} \quad (8)$$

where residualized wages in commuting zone c , for firm i at time τ are denoted by $\text{residual wages}_{i,c,\tau}$ and vacancies of firm i in commuting zone c at time τ is denoted by $v_{i,c,\tau}$

Dependent variables. As a dependent variable, we use log-vacancies and deviations of wages from local average to more closely track the difference of particular establishments from its labor market competitors in the local labor market, the latter distinction, however, does not play a role once we include commuting zone - time fixed effects.⁹

Our unit of analysis is the firm-commuting zone-time level. We have in total over 15 million firm-commuting zone-quarter observations with a total of over 380,000 firms and just over 700 commuting zones. The average commuting zone has postings from 22,000 firms, although this is unevenly distributed. The average firm posts in 170 commuting zones (see [Table 1](#) for further details).

3.2 Monetary Policy shocks

The baseline measure of monetary policy shocks we use is that developed in [Jarociński and Karadi \(2020\)](#) — JK 2020 henceforth. They focus on interest rate surprises in three-month fed funds futures, which exchange a constant interest for the average federal funds rate over the course of the third calendar month in the contract. As regular FOMC

⁹Average regional wages are subtracted to abstract from compositional effects, for example, after a contractionary monetary shock, postings in high wage regions (low labor market power) fall by less, biasing the response of wages up. Subtracting average regional wages allows partialing out commuting zone time characteristics.

meetings are 6 weeks apart from each other, the three-month future reflects the shift in the expected federal funds rate after the following policy meeting. These shocks do not capture surprises to the balance sheet, implicitly assuming that such changes are orthogonal to surprises to the policy rate (and that balance sheet measures would not affect the 3-month futures).

We prefer JK 2020 shocks because they separate pure monetary policy shocks from signaling shocks related to the state of the economy — so called “Fed information” (Romer and Romer, 2000; Nakamura and Steinsson, 2018) or “Fed reacting to information” (Bauer and Swanson, 2023) shocks. The first reflects the fact that economic agents take Fed actions as a signal about the state of the economy and adjust their expectations accordingly. For instance, a surprising monetary loosening can be taken as a sign that the economy is performing poorly and as a result economic agents might, for instance, reduce investment. The “Fed reacting to information” interpretation of the same loosening is that the counter-intuitive response to monetary policy results from the failure to control for recent bad news on the economy known to both the Fed and agents. The effect of Fed information shocks, therefore, goes in the opposite direction to that of monetary policy, and mixing the two together can significantly bias the results and confound channels. As a baseline, we are only interested in the effect of the monetary policy shock and we use the information shock as a control.¹⁰

Appendix Table A7 and Table A8 show that our main results are robust to using other measures of monetary policy shocks, including those of Nakamura and Steinsson (2018), Jarocinski (2021), and Bu et al. (2021).¹¹

¹⁰Appendix Figure A1 presents the time-series of JK 2020 shocks, both monetary policy and central bank information. Monetary policy shocks exhibit both significant tightening and loosening. Our vacancy dataset includes years 2007 and 2010-2019. Over this period, the largest tightening and loosening shocks happened in 2007. The global financial crisis-related loosening cycle started in August 2007 and in the run up to it uncertainty over the state of the economy and thus over monetary policy was elevated.

¹¹Nakamura and Steinsson (2018) use principle component analysis to combine in one shock surprises across the yield curve from one-month to two-year. Jarocinski (2021) estimates four different shocks, including the standard monetary policy shock and three orthogonal shocks that do not affect near-term fed funds futures. The other shocks include: (i) an Odyssean forward guidance shock (a commitment to a future course of policy rates); (ii) a shock to longer-term treasury yields mostly affected by asset purchase announcements; and (iii) a Delphic forward guidance shock (Campbell et al., 2012), which captures the stance of future monetary policy in the sense of a prediction of the appropriate stance of policy, rather than its commitment. Bu et al. (2021) include unconventional policy constructed through a Fama-MacBeth two-step procedure to extract monetary policy shocks from the common component of outcome variables. They conclude that their measure does not contain a significant central bank information effect.

4 Empirical Results

The model in section 2 provides two tests: (1) on equilibrium vacancies and wages; and (2) on responses of vacancies and wages to monetary policy shocks. In this section, we use data on vacancies, wages, and monetary policy shocks (as exogenous demand-shifters) to test which is the dominant source of labor market power across firms — classical monopsony or search monopsony. We find that: (1) in the cross-section, firms that post more vacancies in a regional labor market offer lower wages even after controlling for many observed and unobserved characteristics; and (2) firms that account for more vacancies and post lower wages in a regional labor market raise vacancies by more following monetary policy loosening, without having to increase wages. Both findings are inconsistent with the predictions from a model with classical labor market power only and suggest that search monopsony may be the dominant source of variation in labor market power in the data.

4.1 Test 1: Equilibrium Vacancies and Wages

Test 1: *In equilibrium, classical monopsony implies a positive relationship between vacancies and wages, while search monopsony implies a negative relationship.*

Classical monopsonists curtail vacancies and employment because they internalize that all workers are paid more when the firm offers higher wages to attract new hires. This creates higher markdowns on their wages, implying a positive correlation between vacancies and wages (Proposition 1). In contrast, search monopsonists post more vacancies allowing them to reduce wages, as workers value the higher employment probability, implying a negative correlation between vacancies and wages (Proposition 2). This fundamental dichotomy suggests that studying the cross-sectional correlation of wages and vacancies in the data can provide preliminary evidence on the importance of search market power relative to classical monopsony.

Indeed, we find that a firm’s share of vacancies in a given commuting zone is correlated with *lower* posted wages, suggesting an important presence of search monopsony. [Figure 1](#) plots firms’ commuting zone-level average posted wage on the y-axis against the vacancy share of that same firm in the same commuting zone on the x-axis. To account for the right-skewed nature of vacancy shares, the x-axis uses a log-scale. The left panel plots the relationship for non-college vacancies and the right panel for college vacancies. A large portion of the distribution of vacancy shares offers similar posted wages. In particular, firms with a vacancy share between 0 and 0.00005 (0.005%) post average wages between US\$50,000 and US\$48,000 for non-college vacancies. These firms have extremely low labor market power, and within that group differences are not meaningful. A firm that accounts for a vacancy share of 0.00005 posts 1 out of 20,000 vacancies. However, once a

firm starts to control more than 0.005 % of the market their posted wage declines strongly. For instance, a firm with a vacancy share of 0.1%, posting 1 in every 1000 vacancies, posts an average wage of less than US\$45,000 for non-college vacancies. The same pattern is visible for college vacancies, with posted wages declining more linearly than for non-college vacancies, but with a large drop in posted wages at vacancy shares of about 0.005%. As expected, the overall level of wages posted is significantly higher at around US\$76,000 for firms with a vacancy share of < 0.00005) and around US\$70,000 for firms with a vacancy share of > 0.001 .

The negative relationship between vacancies and wages could be spurious or suffer from compositional biases. For instance, if firms with high vacancy shares hire less-skilled workers, lower wages would not be directly due to their higher vacancy share. The split between college vs. non-college vacancies partly addresses this compositional issue, making only a within college/non-college comparison, but cannot fully dismiss compositional issues, as even within each level of educational attainment, skill requirements and productivity can differ significantly.

We test the importance of search monopsony more formally by estimating vacancy-level regressions that control for a large set of vacancy characteristics, such as commuting zone - time and ONET occupation - time fixed effects. We find that even after controlling for observed and unobserved vacancy, firm, and region characteristics, firms with higher vacancy shares post lower wages on their vacancies (Table 2). We interpret this result as evidence against the model with only classical market power and in favor of the model with search labor market power. We note that any confounding factors of firm-level productivity that may be left in these regressions would bias our estimates towards finding a positive correlation between vacancies and wages.

Using this result that vacancies and wages are negatively correlated in the cross section of firms suggests a back of the envelope calculation that allows to bound the relative importance of search vs classical monopsony. Take an approximation of the cross-firm variation of markdowns (defined in equation 3) around average α and γ : $\text{var}(\nu_i) = \frac{1}{(1+\gamma)^2} \text{var}(\alpha_i) + \frac{(1-\alpha)^2}{(1+\gamma)^4} \text{var}(\gamma_i) + 2\frac{1}{1+\gamma} \frac{1-\alpha}{(1+\gamma)^2} \text{cov}(\alpha_i, \gamma_i)$. Assuming that $\text{cov}(\alpha_i, \gamma_i) = 0$, we divide the contribution coming from the variance in α_i by the variance of the markdown to obtain the share of the variance of markdowns due to search labor market power. After manipulation and substitution, the share coming from the variance of α is given by: $\frac{1}{1+\nu^2} \frac{\text{var}(\gamma_i)}{\text{var}(\alpha)}$. The ratio of variances of search and classical monopsony parameters can be obtained from the approximate formula for the covariance between equilibrium vacancies and wages: $\frac{\text{var}(\gamma)}{\text{var}(\alpha)} = \frac{\text{cov}(v,w)}{v'_\gamma w'_\gamma \text{var}(\alpha)} + \frac{-v'_\alpha w'_\alpha}{v'_\gamma w'_\gamma}$. Given our finding that the covariance between equilibrium vacancies and wages is negative, we obtain: $\frac{\text{var}(\gamma)}{\text{var}(\alpha)} < \frac{-v'_\alpha w'_\alpha}{v'_\gamma w'_\gamma}$.

We then use estimates for the elasticity of search from [Marinescu and Wolthoff \(2020\)](#),

the range of estimates for the average markdown from [Yeh et al. \(2022\)](#), the relationship between parameters α , γ and vacancies and wages from Propositions 1 and 2 to obtain a lower bound of that statistic.¹² We find that at least 55% of markdown variation is driven by the variation in search labor market power.

4.2 Test 2: Response to Monetary Policy Shocks

Model Predictions and Empirical Approach

Test 2: *If wages of all firms respond homogeneously to monetary policy and vacancies respond more in firms with higher equilibrium vacancies and lower equilibrium wages, the model where firms vary only in their extent of classical monopsony can be rejected. This implies that search monopsony drives the response of wages and vacancies to monetary policy.*

To implement Test 2, we run the following two specifications:

$$\log \text{vacancies}_{i,c,t} = \beta^{vv} \text{MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} + \Theta^{vv} \text{controls} + \epsilon_{i,c,t}^{vv} \quad (9)$$

$$\log \text{wages}_{i,c,t} = \beta^{vw} \text{MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} + \Theta^{vw} \text{controls} + \epsilon_{i,c,t}^{vw} \quad (10)$$

where i is a firm, in a local labor market c at time t .¹³

Recall from Proposition 1, classical monosponists offer lower equilibrium vacancies and lower wages. Instead, Proposition 2 suggests search monopsonists offer *higher* equilibrium vacancies and lower wages. Putting these together with Proposition 3 implies that ignoring differences in search monopsony across firms, it should be that we find $\beta^{vv} = \beta^{vw} = 0$, as firms with classical monopsony power should not react differentially to monetary policy shocks. If instead, we allow for heterogeneity in search, proposition 3 states firms with higher equilibrium vacancies and *lower* equilibrium wages (more labor market power) post more vacancies in response to a monetary policy shock. Hence, we should find that $\beta^{vv} > 0$, but $\beta^{vw} < 0$. This is the essence of Test 2.

In addition, Test 2 requires that wages do not respond differentially to either classical or search market power. If that were not the case, estimates of β^{vv} and β^{vw} in equations (9) and (10) could reflect non-labor market characteristics that vary across firms, crucially firm productivity rather than labor market power. To assure that the conditions for Test

¹²See Appendix 21 and 21 for more details on the relationship between α , γ and vacancies and wages

¹³**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies or wages, respectively, (ii) equilibrium vacancies or wages depending on whether equation (9) or (10) are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, and (iv) commuting zone-time effects that absorb any time-varying regional shocks, such as local demand shocks.

2 hold, we run the following two additional regressions:

$$\log wages_{i,c,t} = \beta^{vv} \text{MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} + \Theta^{vv} \text{controls} + \epsilon_{i,c,t}^{vv} \quad (11)$$

$$\log wages_{i,c,t} = \beta^{ww} \text{MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} + \Theta^{ww} \text{controls} + \epsilon_{i,c,t}^{ww} \quad (12)$$

where again i is a firm, in a local labor market c at time t .¹⁴

Test 2 requires the interaction term between the monetary policy shock and equilibrium characteristics to be zero, $\beta^{vv} = \beta^{ww} = 0$.

Main Results

Table 3 shows estimates for β^{vv} , β^{vw} , β^{wv} , and β^{ww} as defined in equations (9)-(12).

In a model with search, labor market power amplifies the effect of monetary policy on vacancies. Instead, in a model with classical labor market power, the response of vacancies would not vary.

The first column estimates a positive and significant β^{vv} . In response to an accommodative (contractionary) monetary policy shock, firms that account for a larger share of regional vacancies increase (decrease) their vacancy postings by more than their counterparts. This result is consistent with firms with higher search labor market power responding more to monetary policy shocks. Firms with higher classical labor market power would have lower equilibrium vacancy shares and their differential vacancy response would be muted.

Column (2) of Table 3 estimates β^{vw} , or whether firms that pay relatively lower wages compared to their counterparts react differently to monetary policy in terms of their vacancy postings. Under both classical and search labor market power, lower wages are associated with more labor market power. β^{vw} is estimated to be negative and significant, meaning that firms that pay *lower* wages also increase vacancy postings more in response to monetary policy easing.

These estimates strongly suggest that heterogeneity in search labor market power is a key driver of vacancies and not classical labor market power, which would predict a null result in both columns (1) and (2).¹⁵

Finally, columns (3) and (4) of Table 3 present estimates for equations (11) and (12). We cannot reject that β^{wv} , and β^{ww} are both zero, i.e. we cannot find a differential wage response to monetary policy of firms with either higher share of vacancies or higher

¹⁴**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies and wages, (ii) equilibrium vacancies or wages depending on whether equation (11) or (12) are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, and (iv) commuting zone-time effects that absorb any time-varying regional shocks, such as local demand shocks.

¹⁵Note that Lightcast data for posted wages is much less comprehensive than data on vacancies and that explains the drop in observations between column 1 and 2.

wages. These support our conclusion that columns (1) and (2) are driven by heterogeneity in search monopsony power rather than other potential non-labor confounding factors.

Alternative assumptions

[Table 4](#) re-estimates column (1) of [Table 3](#) under alternative assumptions. In column (1) we include firm fixed effects to control for unobserved and observed time-invariant heterogeneity at the firm-level, for instance, the average number of vacancies a firm posted during our sample period. It also includes commuting zone fixed effects to control for time-invariant heterogeneity at the commuting zone level. Note that the exclusion of any type of time fixed effects allows us to identify directly the effect of monetary policy shocks, which is collinear in a specification once time fixed effects are included. As expected, monetary policy easings are associated with more vacancy postings. Quantitatively, a 10 bp monetary policy easing is associated 7% higher vacancy postings. This effect is amplified when firms have a larger share of vacancies, as already shown in column (1) of [Table 3](#) under a different set of fixed effects. Column (2) replaces commuting zone fixed effects with time fixed effects and column (3) has firm, commuting zone, and time fixed effects. Column (4) shows in comparison our baseline specification and column (5) introduces firm-time fixed effects, which allows comparing vacancy posting within firms across commuting zones. The interaction coefficient is always positive and statistically significant, indicating that firms with higher vacancy shares are more responsive in terms of their vacancy postings than their counterparts. Quantitatively, the interaction coefficient is relatively stable varying between 5.4 and 7.9, implying that firms with a 1 percentage point higher share of regional vacancies have a between 0.54 and 0.79 pp stronger response to a 10 bp monetary policy shock. For instance, a 10 bp accommodative monetary policy shock would increase vacancy postings by 7% for a firm without any vacancy postings, while for a firm that has 1% vacancy share (or the 84th percentile in the vacancy-weighted distribution of vacancy shares) the response is between 7.53% and 7.79%.

[Table 5](#) re-estimates column (2) of [Table 3](#) under alternative assumptions. The first column estimates the specifications only with firm and commuting zone fixed effect. The effect of monetary policy shocks on vacancy postings is amplified for firms that pay low wages, consistent with column (2) of [Table 3](#). Column (2) replaces commuting zone fixed effects with time fixed effects and column (3) has firm, commuting zone, and time fixed effects. Column (4) shows in comparison our baseline specification and column (5) introduces firm-time fixed effects, which allows comparing vacancy posting within firms across commuting zones. The interaction coefficient is always negative and statistically significant, except for column (5), indicating that firms with lower wages are more responsive in terms of their vacancy postings than their counterparts. One potential explanation

for why the coefficient becomes insignificant in column (5) is that some firms set wages nationally, as pointed by [Hazell et al. \(2021\)](#). In this case, firm-time fixed effects would absorb any wage variation, rendering the estimated coefficient insignificant and very close to zero. This mechanical feature, however, does not influence the results of the regressions with the vacancy share interaction, as vacancies vary across a firm’s locations. National wage-setting would only weaken the mechanism we highlight, but not undo it. Except for the last column, quantitatively, the interaction coefficient is relatively stable varying between -0.058 and -0.067, implying that firms with one standard deviation lower wages have between 0.58 and 0.67 pp stronger response to a 10 bp monetary policy shock. For instance, a 10 bp accommodative monetary policy shock would increase vacancy postings by 8.7% for the average firm, while for a firm that has one standard deviation lower wages the response is between 9.28% and 9.37%.

[Table 6](#) re-estimates column (3) of [Table 3](#) under alternative assumptions. The first column estimates the specifications only with firm and commuting zone fixed effects. This allows us to identify directly the effect of monetary policy shocks on wages, which is collinear in a specification once time fixed effects are included. As expected, monetary policy easings are associated with higher wage growth. Quantitatively, a 10 bp monetary policy easing is associated 1.5% higher wage growth. This effect is relatively stable across firms with varying degree of vacancy shares, as the model with search labor market would predict, especially as we move across columns that successively include a more extensive set of fixed effects in the regressions. Those results are also visible in [Table 7](#), where firms that are paying a varying degree of wages are also not responding differently in terms of wage setting in response to monetary policy shocks.

[Table 8](#) and [Table 9](#) re-estimate columns (2) and columns (4) of [Table 3](#) using different definitions of wages. As defined in equation (8) in section 3, equilibrium wages are measured as residuals from occupation=commuting zone level regressions and then standardized, averaged, and cumulated. For robustness, we use the pure wage measure in column (2), with ONET-time and commuting zone-time fixed effects in column (3), and ONET-time and commuting zone fixed effects in column (4). In both regression tables the results are robust to the way we measure wages. In [Table 8](#) the wage coefficient is always negative and statistically significant, while in [Table 9](#) the coefficients are statistically insignificant, as predicted by the model with search labor market power.

Local Projections

So far, we have only analyzed the contemporaneous effect of monetary policy. Monetary policy shocks are known to take time to affect macroeconomic variables. We employ local projection (LP) methods to confirm the finding that search monopsony is a key feature

of the data.

We estimate the following LP versions of equations (9)-(12):

$$\frac{1}{H} \sum_{h=0}^H Y_{i,c,t+h} = \alpha_H + \beta_H^{YX} \text{MP easing}_t \times X_{i,c,t-1} + \Theta^{YX} \text{controls} + \varepsilon_{i,c,t+H}^{YX} \quad (13)$$

where $Y = \log \text{vacancies}$ or $\log \text{wages}$, $X = \text{equilibrium vacancies}$ or equilibrium wages , H is a given quarterly horizon, and i is a firm in a local labor market c at time t .^{16,17}

Figure 4 shows the point estimates of β_H^{YX} from equation (13) along with 95 percent confidence intervals allowing for firm and commuting-zone double clustering. The top left panel displays the interaction coefficient β^{vv} . As in the static regressions, firms with a higher share of equilibrium vacancies respond by posting more (less) vacancies following expansionary (contractionary) monetary policy, as indicated by the positive coefficient. The coefficient remains positive and statistically significant for 6 quarters after the monetary policy shock occurred.

The bottom left panel plots the interaction coefficient β^{vw} . The coefficient is persistently negative, indicating that firms that pay lower wages increase their vacancy postings by more relative to the counterparts.

The two right panels show β^{vv} (top) and β^{ww} (bottom) or the responses of wages to expansionary monetary policy shocks depending on equilibrium vacancy shares and wages, respectively. As was the case in the static regressions, these responses are economically and statistically insignificant within a six-quarter horizon.

4.3 Heterogeneity across different types of jobs

In what follows, we investigate the effects of labor market power across different types of job postings. Presumably, low-skilled workers exhibit lower regional mobility, while skilled workers are likely to move to regions with higher wage premiums (Dahl, 2022). This creates a locked-in effect for low-skilled workers, making them more dependent on local labor market conditions. Moreover, low-skilled workers are disproportionately affected by automation, which further reduces their bargaining power (Acemoglu and Restrepo, 2022; Acemoglu et al., 2021). This suggests testing our predictions across more and less skilled workers to assess whether the importance of search monopsony also relates to workers

¹⁶**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies and wages, (ii) equilibrium vacancies or wages depending on whether $X = \text{equilibrium vacancies}$ or equilibrium wages are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, and (iv) commuting zone-time effects that absorb any time-varying regional shocks, such as local demand shocks.

¹⁷For these regressions, if Lightcast data does not report a vacancy for a given firm in a given commuting zone and quarter we assume there were no vacancies posted.

skill levels. Lightcast provides granular data on postings, including on skill and education requirements. We focus on two types of requirements. First, we differentiate between college vs. non-college vacancies. In our sample, $\approx 40\%$ of vacancies require college education. We also study the degree of tech-savviness of vacancies, by differentiating between vacancies that require software skills and those that do not, in the spirit of [Acemoglu et al. \(2021\)](#) who use Lightcast data to identify AI vacancies. Vacancies that require software skills make up $\approx 28\%$ of all vacancies. As one would expect, college vacancies and tech-savvy vacancies are strongly related to each other, with a correlation between the two vacancy types of $\approx 29\%$.

To test the model predictions across these types, we run the following versions of equations (9)-(10):

$$\begin{aligned} \log \text{vacancies}_{i,c,t,j} = & \beta^v \text{MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} + \\ & \gamma^v \text{MP easing}_t \times \text{equilibrium vacancies}_{i,c,t-1} \times \text{Type}_j + \Theta^v \text{controls} + \varepsilon_{i,c,t}^v \end{aligned} \quad (14)$$

$$\begin{aligned} \log \text{vacancies}_{i,c,t,j} = & \beta^w \text{MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} + \\ & \gamma^w \text{MP easing}_t \times \text{equilibrium wages}_{i,c,t-1} \times \text{Type}_j + \Theta^w \text{controls} + \varepsilon_{i,c,t}^w \end{aligned} \quad (15)$$

where Type_j is a dummy that takes the value of 0 or 1 depending on the characteristic of the job posting, and i is a firm in a local labor market c at time t .¹⁸ The triple interaction coefficient γ captures whether there is significant heterogeneity across a particular type of vacancy. If the double interaction (β) has a different sign than the triple interaction (γ), that would mean that the effect is weaker for $[\text{Type} = 1]$.

[Table 10](#) shows estimates of [Equation 14](#). Column 1 confirms our main result that vacancies respond more (less) to monetary loosening (tightening) when a firm controls a greater share of vacancies. But does this effect depend on they type of vacancy? The interaction between the monetary policy shock and the vacancy type dummy is negative and statistically significant in column 1, where the vacancy type dummy is one if the vacancy is a college vacancy. This implies that college vacancies respond less strongly to monetary policy than non-college vacancies.

Column 2 illustrates whether this effect is partly driven by labor market power. Indeed, the interaction between the monetary policy shock, vacancy share and the vacancy type

¹⁸**controls** include: (i) the Federal Reserve information shock and its interactions with equilibrium vacancies or wages, respectively, (ii) equilibrium vacancies or wages depending on whether equation (14) or (15) are estimated, (iii) firm fixed effects that absorb any time-invariant firm variation, (iv) commuting zone-time effects that absorb any time-varying regional shocks, such as local demand shocks, and (v) type-time fixed effects that absorb any time-varying type-specific changes, such as shifts technology benefiting skilled workers.

dummy is positive and statistically significant. The positive triple interaction term shows that labor demand effects of labor market power in response to a monetary policy shock are stronger for non-college vacancies. When interpreting the economic magnitude, we can see that the effect is around -7.8 for non-college vacancies and $(-7.8 + 2.9) = -4.9$ for college vacancies.

The results are similar although weaker for software-related vacancies. In column 3 we can see that software vacancies are less responsive to monetary policy in general, and when firms exhibit labor market power their adjustment seems to be done along the non-software dimension, rather than on the more tech-related vacancies.

Looking at the full dynamics of effects over vacancy types, we find that the effect of labor market power for vacancies not requiring college only grows over time and that of vacancies requiring software skills also becomes stronger (see ?? and ??).

?? explicitly compares the response of labor demand over time for the median firm in terms of labor market power to a firm with a high degree of labor market power (95th percentile) separating the responses of college (left) and non-college vacancies (right). In the case of non-college vacancies, the response of vacancies to a monetary policy shock of firms with labor market power is persistently different and increases over time compared to those that do not have labor market power. For college vacancies there isn't a meaningful difference beyond the first quarter.

Our specification is based on symmetric effects of positive and negative monetary policy surprises.¹⁹ Following a contractionary (expansionary) surprise shock, firms with labor market power cut (expand) their vacancies by more than firms without labor market power, although the effect of the monetary policy shock on wages is similar across firms with and without labor market power. However, there are important compositional effects. Following a contractionary (expansionary) shock, the share of vacancies by high market power firms decreases (increases) and since these firms pay lower wages on average and have a higher markdown, this dampens the effect of the monetary policy shock on aggregate wages.

5 Conclusion

Where does labor market power come from and why does it matter? A large body of work has focused on monopsony power that derives from imperfect substitution between labor markets, for example because jobs are closer to where workers live. Under this type of “classical” monopsony, wages are driven below the marginal revenue of labor

¹⁹We check for asymmetries in the effects but found no significant evidence of differential effects of positive and negative surprise shocks. Results are available from the authors upon request

inefficiently and employment is too low. That deadweight loss creates a rationale for several policies, like minimum wages. Another strand of the literature studies a different type of monopsony power that arises due to search frictions. Because applicants face uncertain job prospects and that uncertainty is unwarranted, firms that are better at converting applicants into employees derive an advantage that can be seen as monopsony power too. Unlike classical monopsony, such “search” monopsony can be efficient or even if inefficient, generate excess employment. Minimum wages may be less compelling if search monopsony is pervasive.

To the best of our knowledge this paper is the first to include the two types of monopsony power in a simple framework and to test their relative pervasiveness. To do so, we first derive predictions for equilibrium vacancies and wages and responses of vacancies and wages to demand shocks in a model that is rich in “classical” monopsony with simplified directed search elements. These predictions draw a sharp distinction between classical and search monopsony.

We test these predictions by using the near universe of online vacancy postings in the U.S. and find that the cross-section of wages and vacancies is better rationalized by the presence of search monopsony, implying a back-of-the-envelope calculation that at least 55% of markdown cross-firm variation is driven by cross-firm variation in search labor market power. Moreover, we also find that the response of vacancies and wages to demand shocks measured by exogenous monetary policy shocks can only be explained with a preponderance of search rather than classical monopsony power. We hope that our paper will spur further quantitative research in distinguishing these two forms of monopsony power and the optimality of labor market policies, such as minimum wages.

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Table 1: Summary of the Panel Structure

	Total	Firms	Commuting Zone	Time
Number of Observations	15,810,352	387,107	708	43
Average Number of Firms	387,107	-	22,412	103,230
Average Number of CZ	708	170	-	704
Average Number of Periods	43	29	42	-

This table reports the total number of observations and the number of observations across the time and geographical dimensions of the data

Table 2: Equilibrium Wages and Vacancies (Test 1)

	Log wage _{<i>v,i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
Equilibrium Vacancies	-0.232*** (-3.84)	-0.232*** (-3.84)	-0.175*** (-3.91)	-0.188*** (-3.49)	-0.193*** (-4.14)
College _{<i>v,i,c,t</i>}	0.238*** (41.47)	0.238*** (41.47)	0.172*** (42.51)	0.170*** (43.17)	0.162*** (45.50)
Software Skills _{<i>v,i,c,t</i>}	0.028*** (8.13)	0.028*** (8.13)	0.016*** (5.09)	0.015*** (5.01)	0.014*** (5.08)
Specialized _{<i>v,i,c,t</i>}	0.088*** (29.75)	0.088*** (29.75)	0.065*** (25.62)	0.065*** (27.11)	0.063*** (28.45)
Routine Manual _{<i>v,i,c,t</i>}	-0.110*** (-18.86)	-0.110*** (-18.86)	0.038*** (4.20)	0.038*** (4.31)	0.050*** (5.97)
Routine Cognitive _{<i>v,i,c,t</i>}	-0.144*** (-28.17)	-0.144*** (-28.17)	0.061*** (6.99)	0.062*** (7.72)	0.077*** (9.46)
Non-Routine Manual Physical _{<i>v,i,c,t</i>}	-0.057*** (-9.41)	-0.057*** (-9.41)	0.040*** (4.48)	0.045*** (5.12)	0.039*** (5.27)
Non-Routine Manual Inter-Personal _{<i>v,i,c,t</i>}	0.045*** (10.42)	0.045*** (10.42)	0.035*** (4.81)	0.035*** (4.91)	0.039*** (5.01)
Non-Routine Cognitive Analytical _{<i>v,i,c,t</i>}	0.064*** (12.24)	0.064*** (12.24)	0.111*** (16.58)	0.113*** (17.61)	0.112*** (16.69)
Non-Routine Cognitive Personal _{<i>v,i,c,t</i>}	0.159*** (33.83)	0.159*** (33.83)	0.045*** (8.67)	0.046*** (9.32)	0.046*** (10.02)
Obs.	12,737,416	12,737,416	12,237,549	12,233,118	12,232,649
Firm FE	✓	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓	✓
Industry × Time FE	✓	✓	✓	✓	✓
ONET × Time FE			✓	✓	✓
ONET × CZ FE				✓	✓
ONET × Industry FE					✓
No. Firms	174,270	174,270	171,560	171,524	171,521

This table reports results for the following vacancy-level regression: $\text{Log wage}_{v,i,c,t} = \alpha + \beta \text{Equilibrium Vacancies}_{i,c,t} + \theta X_{v,i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \gamma_{ind,t} + \gamma_{occ,t} + \gamma_{occ,c} + \gamma_{occ,ind} + \varepsilon_{v,i,c,t}$, where $\text{Log wages}_{v,i,c,t}$ is defined as the log of posted wage in vacancy v for firm i , in commuting zone c in quarter t . $\text{Equilibrium Vacancies}_{i,c,t}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter t . $X_{v,i,c,t}$ is a vector of vacancy characteristics as defined in [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time, and $\gamma_{ind,t}$ are industry-time, $\gamma_{occ,t}$ are occupation-time, $\gamma_{occ,c}$ are occupation-CZ, and $\gamma_{occ,ind}$ are occupation-industry fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Monetary Policy under Labor Market Power (Test 2)

	Log Vacancies _{<i>i,c,t</i>}		Log Wages _{<i>i,c,t</i>}	
	(1)	(2)	(3)	(4)
MP easing _{<i>t</i>} × Equilibrium Vacancies	6.906** (2.707)		0.128 (0.377)	
MP easing _{<i>t</i>} × Equilibrium Wages		-0.061** (0.028)		-0.020 (0.033)
Obs.	15,069,930	4,135,037	3,545,581	1,827,037
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
No. Firms	355,254	145,726	216,310	110,303

This table reports estimates of the following regression: $Y_{i,c,t} = \alpha + \beta \text{MP easing}_t \times X_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_i + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $Y_{i,c,t}$ is the dependent variable (Log Vacancies_{*i,c,t*}, Log Wages_{*i,c,t*}). Log Vacancies_{*i,c,t*} is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . Log Wages_{*i,c,t*} is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_{*t*} is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $X_{i,c,t}$ is the equilibrium wage or equilibrium vacancy measure. Equilibrium vacancy measure is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t - 1$. For more details see section 3. Equilibrium wage measure is defined as the standardized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. γ_i is firm-level fixed effects, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 4: Labor Demand Effect of Monetary Policy - Vacancies to Vacancies

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.696*** (0.035)				
Equilibrium Vacancies	20.318*** (1.536)	14.958*** (1.277)	20.866*** (1.563)	22.265*** (1.569)	22.713*** (1.655)
MP easing _{<i>t</i>} × Equilibrium Vacancies	5.442** (2.329)	5.439*** (1.833)	7.624*** (2.397)	6.906** (2.711)	7.895** (3.862)
Obs.	15,070,026	15,070,026	15,070,026	15,069,930	12,851,727
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	355,254	355,254	355,254	355,254	199,839

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where Log Vacancies_{*i,c,t*} is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP_{*t*} is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. Eq.Vac_{*i,c,t-1*} is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t - 1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 5: Labor Demand Effect of Monetary Policy - Vacancies to Wages

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.874*** (0.057)				
Equilibrium Wages	0.015*** (0.004)	-0.004 (0.005)	0.013*** (0.004)	0.013*** (0.004)	0.013*** (0.004)
MP easing _{<i>t</i>} × Equilibrium Wages	-0.058** (0.028)	-0.067** (0.027)	-0.062** (0.028)	-0.061** (0.028)	0.008 (0.025)
Obs.	4,135,855	4,135,856	4,135,855	4,135,037	3,295,071
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	145,726	145,726	145,726	145,721	47,267

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq. Wage}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq. Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 6: Labor Demand Effect of Monetary Policy - Wages to Vacancies

	Log Wages _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.148*** (0.024)				
Equilibrium Vacancies	0.056 (0.061)	-0.011 (0.093)	0.112* (0.065)	0.108 (0.077)	0.390*** (0.081)
MP easing _{<i>t</i>} × Equilibrium Vacancies	-0.495* (0.279)	0.009 (0.271)	0.090 (0.277)	0.128 (0.378)	0.363 (0.485)
Obs.	3,546,366	3,546,366	3,546,366	3,545,581	2,715,673
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	216,315	216,315	216,315	216,310	97,856

This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq. Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq. Vac}_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t - 1$. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 7: Labor Demand Effect of Monetary Policy - Wages to Wages

	Log Wages _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.161*** (0.027)				
Equilibrium Wages	0.134*** (0.004)	0.136*** (0.004)	0.133*** (0.004)	0.134*** (0.004)	0.123*** (0.003)
MP easing _{<i>t</i>} × Equilibrium Wages	-0.021 (0.032)	-0.012 (0.032)	-0.013 (0.033)	-0.020 (0.033)	0.016 (0.031)
Obs.	1,828,640	1,828,641	1,828,640	1,827,037	1,359,325
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	110,318	110,318	110,318	110,303	36,606

This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Labor Demand Effect of Monetary Policy - Vacancies to Wages (other measures)

	Log Vacancies _{<i>i,c,t</i>}			
	(1)	(2)	(3)	(4)
Equilibrium Wages	0.013*** (0.004)	0.054*** (0.001)	0.009** (0.004)	0.014*** (0.004)
MP easing _{<i>t</i>} × Equilibrium Wages	-0.061** (0.028)	-0.126*** (0.017)	-0.056** (0.028)	-0.060** (0.028)
Obs.	4,135,037	6,133,843	4,134,099	4,135,527
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
No. Firms	145,721	188,583	145,712	145,725
Residuals fe				
ONET × Time FE	✓		✓	✓
CZ × Time FE	✓			
CZ FE				✓
CZ × ONET FE			✓	

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Wage}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t . For column 4 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as a cumulative average wage. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9: Labor Demand Effect of Monetary Policy - Wages to Wages (other measures)

	Log Wages _{<i>i,c,t</i>}			
	(1)	(2)	(3)	(4)
Equilibrium Wages	0.134*** (0.004)	0.881*** (0.003)	0.125*** (0.004)	0.135*** (0.004)
MP easing _{<i>t</i>} × Equilibrium Wages	-0.020 (0.033)	-0.007 (0.044)	-0.025 (0.034)	-0.020 (0.033)
Obs.	1,827,037	3,577,157	1,826,927	1,827,072
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
No. Firms	110,303	216,780	110,298	110,303
Residuals fe				
ONET × Time FE	✓		✓	✓
CZ × Time FE	✓			
CZ				✓
CZ × ONET FE			✓	

This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t . For column 4 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as a cumulative average wage. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table 10: Labor Demand Effect of Monetary Policy - Heterogeneity to Vacancies

	Log Vacancies _{<i>i,c,t,j</i>}			
	(1)	(2)	(3)	(4)
Equilibrium Vacancies _{<i>i,c,t</i>}	17.417*** (1.202)	18.600*** (1.270)	17.751*** (1.220)	21.119*** (1.434)
Type	-0.132*** (0.016)		-0.204*** (0.013)	
MP easing _{<i>t</i>} × Equilibrium Vacancies _{<i>i,c,t</i>}	5.056** (1.986)	6.603*** (2.120)	5.144*** (1.922)	6.116** (2.922)
MP easing _{<i>t</i>} × Type	-0.343*** (0.043)		-0.163*** (0.041)	
Equilibrium Vacancies _{<i>i,c,t</i>} × Type		-2.423*** (0.561)		-8.017*** (0.711)
MP easing _{<i>t</i>} × Equilibrium Vacancies _{<i>i,c,t</i>} × Type		-3.305** (1.635)		-3.133 (2.527)
Obs.	19,007,773	19,007,773	17,956,074	17,956,074
Vacancy Type	college	college	software	software
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
Vac. Type × Time FE		✓		✓

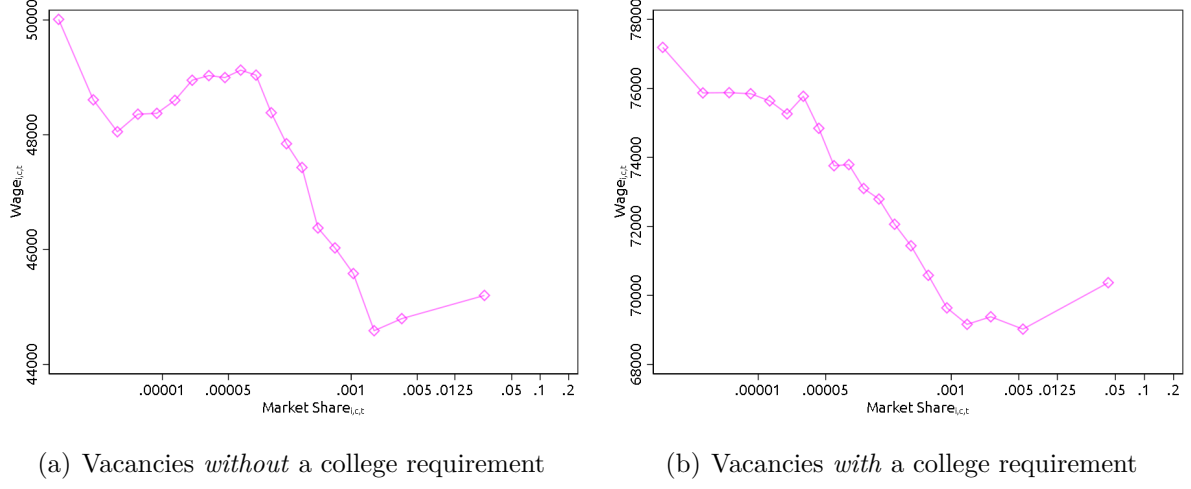
This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t . For column 4 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as a cumulative average wage. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Labor Demand Effect of Monetary Policy - Heterogeneity to Wages

	Log Vacancies _{<i>i,c,t,j</i>}			
	(1)	(2)	(3)	(4)
Equilibrium Wages	0.075*** (0.007)	-0.059*** (0.011)	0.107*** (0.008)	0.020** (0.009)
Type	-0.096*** (0.019)		-0.222*** (0.014)	
MP easing _{<i>t</i>} × Equilibrium Wages	-0.156*** (0.048)	-0.218*** (0.077)	-0.126*** (0.044)	-0.161*** (0.055)
MP easing _{<i>t</i>} × Type	-0.182*** (0.053)		-0.173*** (0.043)	
Equilibrium Wages × Type		0.301*** (0.021)		0.275*** (0.018)
MP easing _{<i>t</i>} × Equilibrium Wages × Type		0.121 (0.086)		0.143* (0.081)
Obs.	11,082,029	11,082,029	10,588,514	10,588,514
Vacancy Type	college	college	software	software
Firm FE	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓
Vac. Type × Time FE		✓		✓

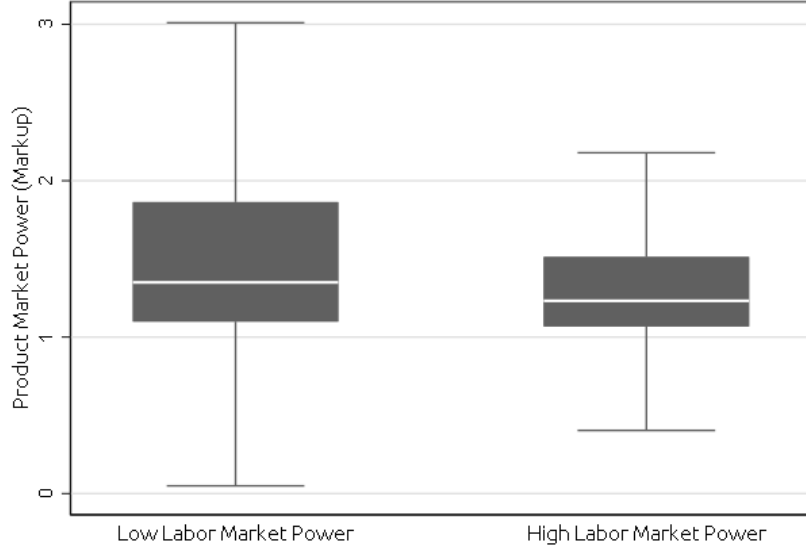
This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the firm i wage in the commuting zone c at the time t from the average wage in the corresponding commuting zone. MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined differently in each column to access robustness. For columns 1, 5 and 6, $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; included fixed effects are, respectively, ONET-time and commuting-zone(CZ)-time, ONET-time and commuting-zone(CZ)-ONET, ONET-time and commuting-zone(CZ). For column 3 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as log of the overtime average salary for the firm i in commuting zone c at the quarter t . For column 4 the $\text{Eq.Wage}_{i,c,t-1}$ is defined as a cumulative average wage. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 1: Vacancy Share and Wages



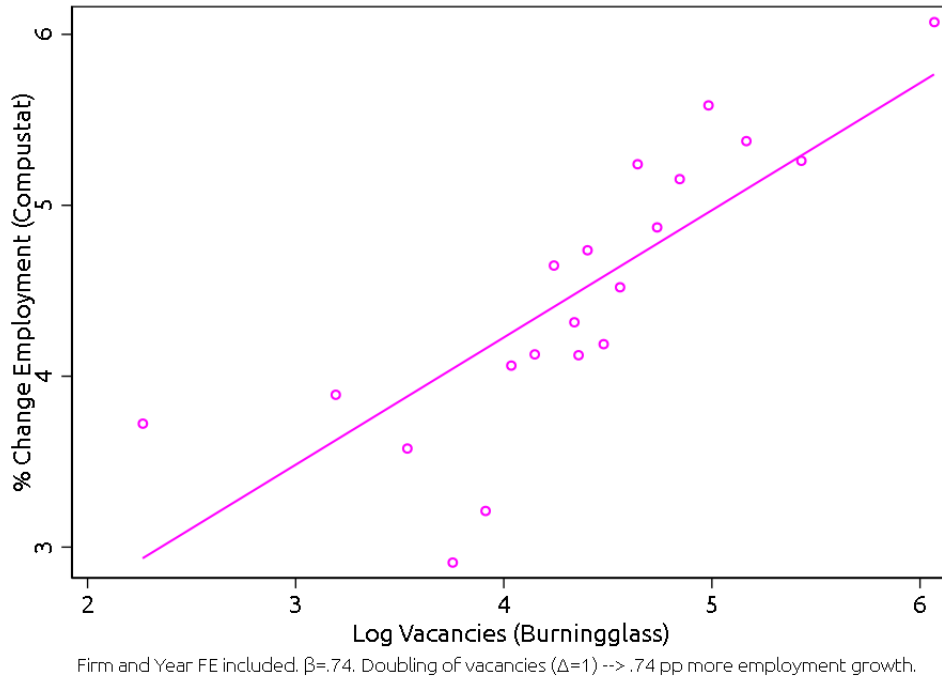
This figure plots a local polynomial smooth of wages on vacancy share for non-college (left panel) and college (right panel) vacancies. The wages are defined as the average wage posted by firm i in commuting zone c in quarter t . The vacancy share defined as $\text{Vacancy Share}_{i,c,t} = \frac{\sum_{\tau \leq t} v_{i,c,\tau}}{\sum_{\tau \leq t} \sum_i v_{i,c,\tau}}$ for each firm i in commuting zone c in quarter t .

Figure 2: Product Market Power and Labor Market Power



This figure plots the distribution of product market power for firms with and without labor market power. Labor market power firms are defined as those that are in the top 5% of labor market power in at least one commuting zone. Product market power is defined as the estimated markup following the approach by [De Loecker et al. \(2020\)](#). The white line reflects the median. The upper (lower) side of the box reflects the 75th (25th) percentile of the markup distribution. The adjacent lines at the top (bottom) reflect the maximum (minimum) value of the markup distribution.

Figure 3: Sensitivity of Employment to Vacancy Postings



This figure plots a binscatterplot between the log change in employment from Compustat on Log Vacancy postings from Lightcast: $\Delta Employment_{i,t} = \alpha_i + \alpha_t + \beta_1 \text{Log Vacancies}_{i,t} + \epsilon_{i,t}$ where $\Delta Employment_{i,t}$ is the log change in employment of firm i between year t and $t - 1$ in Compustat. $\text{Log Vacancies}_{i,t}$ is the log number of vacancies posted by firm i in year t from Lightcast.

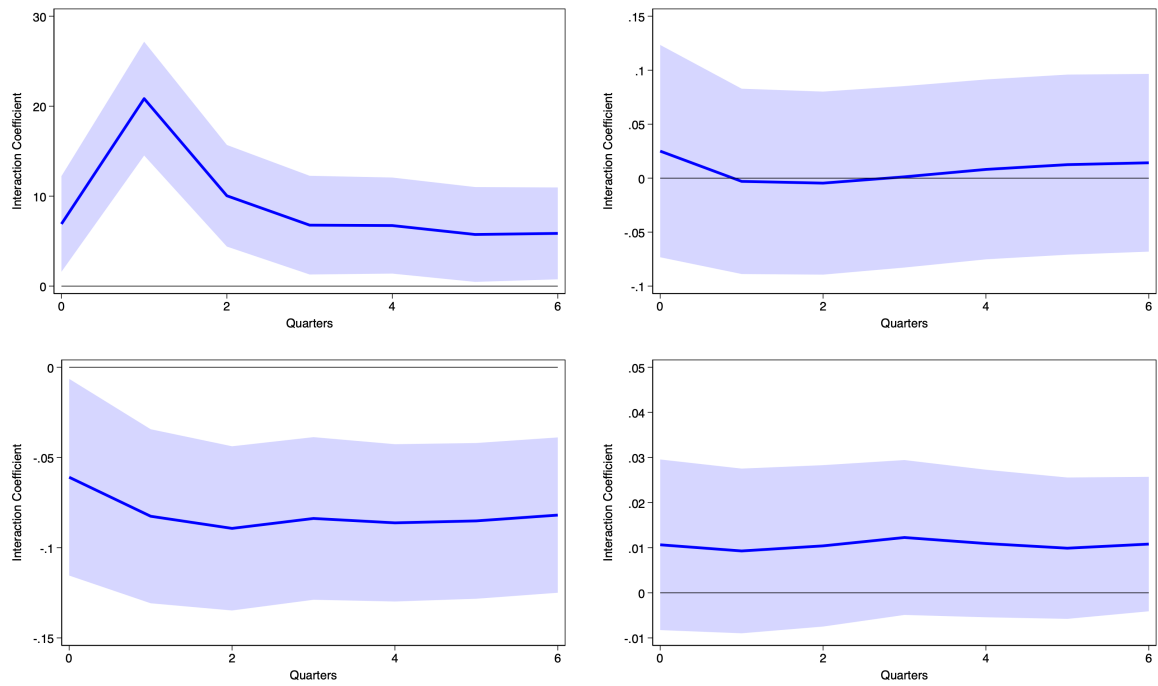
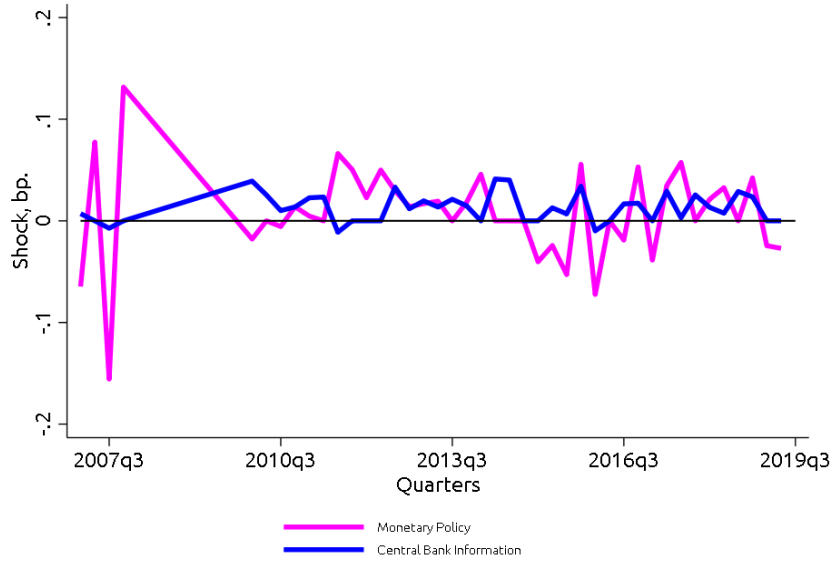


Figure 4: Cumulative Impulse Response of vacancies and wages to monetary policy, interaction coefficient between monetary policy easing shock and equilibrium vacancy and wage measure. Top left: Vacancies to Vacancies, Top right: wages to vacancies, Bottom left: Vacancies to wages, Bottom right: Wages to wages. Please see caption of table 1 for more details.

A Appendix Tables and Figures

Figure A1: Monetary Policy Shocks in Percentage Points (JK 2020)



This figure plots Shocks time series used in the paper. The shock series is developed by [Jarociński and Karadi \(2020\)](#). The positive value of the shock reflects monetary policy easing. The unit is in percentage points. Monetary Policy reflects the shock component that can be assigned to the direct effects of Monetary Policy. Central Bank Information component measures the shock component associated with the effects of the Fed information.

Table A1: Labor Demand Effect of Monetary Policy - Vacancies to Vacancies - Wage Sample

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.666*** (0.055)				
Equilibrium Vacancies _{<i>i,c,t</i>}	16.721*** (1.703)	12.330*** (1.648)	17.171*** (1.733)	19.714*** (1.727)	19.399*** (1.915)
MP easing _{<i>t</i>} × Equilibrium Vacancies _{<i>i,c,t</i>}	6.752*** (2.504)	6.181*** (2.336)	8.694*** (2.607)	10.048*** (2.663)	18.246*** (4.894)
Obs.	3,546,366	3,546,366	3,546,366	3,545,581	2,715,673
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	216,315	216,315	216,315	216,310	97,856

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Vac}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Vac}_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t - 1$. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Labor Demand Effect of Monetary Policy - Vacancies to Wages - Wage Sample

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.727*** (0.070)				
Equilibrium Wages _{<i>i,c,t</i>}	0.023*** (0.007)	-0.007 (0.008)	0.019*** (0.007)	0.019*** (0.007)	0.021*** (0.007)
MP easing _{<i>t</i>} × Equilibrium Wages _{<i>i,c,t</i>}	-0.110** (0.050)	-0.121** (0.048)	-0.087* (0.047)	-0.086* (0.046)	-0.030 (0.048)
Obs.	1,828,640	1,828,641	1,828,640	1,827,037	1,359,325
Firm FE	✓	✓	✓	✓	
CZ FE	✓		✓		
Time FE		✓	✓		
CZ × Time FE				✓	✓
Firm × Time FE					✓
No. Firms	110,318	110,318	110,318	110,303	36,606

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP}_t \times \text{Eq.Wage}_{i,c,t-1} + \theta Z_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $\text{Eq.Wage}_{i,c,t-1}$ is defined as the standartized average cumulative residual of the ONET-level regression. The ONET-level regression is given by $\text{Wage}_{i,c,n,t} = \alpha + \gamma_{n,t} + \gamma_{c,t}$, where Wage is the wage of the firm i in the commuting zone c at the time t for ONET occupation n ; $\gamma_{n,t}$ are ONET-time fixed effects and $\gamma_{c,t}$ are CZ-time fixed effects. For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Labor Demand Effect of Monetary Policy - ONET, Vacancies to Wages, no lag

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	1.156*** (0.345)	1.069*** (0.345)	1.058*** (0.347)		
Equilibrium Wages _{<i>i,c,t</i>}	0.136*** (0.009)	0.086*** (0.006)	0.084*** (0.006)	0.087*** (0.006)	0.085*** (0.006)
MP easing _{<i>t</i>} × Equilibrium Wages _{<i>i,c,t</i>}	-0.077** (0.031)	-0.068** (0.031)	-0.066** (0.032)	-0.062* (0.032)	-0.062* (0.033)
Obs.	17,711,407	17,711,407	17,707,982	17,711,060	17,707,574
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	252,173	252,173	252,148	252,172	252,147

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP easing}_t \times X_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t for the ONET-occupation n . MP easing_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $X_{i,c,n,t-1}$ is defined as the log of the overtime average salary offered by firm i in commuting zone c at quarter $t-1$ for the ONET-occupation n . For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Labor Demand Effect of Monetary Policy - ONET, Vacancies to Vacancies, no lag

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.450*** (0.028)	0.473*** (0.028)	0.471*** (0.027)		
Equilibrium Vacancies	0.774*** (0.043)	1.368*** (0.060)	1.770*** (0.079)	1.461*** (0.062)	1.912*** (0.083)
MP easing _{<i>t</i>} × Equilibrium Vacancies	0.443*** (0.159)	0.617*** (0.157)	0.811*** (0.160)	0.693*** (0.182)	0.906*** (0.187)
Obs.	17,932,913	17,932,913	17,929,754	17,932,607	17,929,414
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	213,614	213,614	213,583	213,612	213,580

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,n,t} = \alpha + \beta \text{MP easing}_t \times X_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,n,t}$, where $\text{Log Vacancies}_{i,c,n,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t for the ONET-occupation n . MP easing_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), and thus a positive value reflects monetary policy easing. $X_{i,c,n,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$ for ONET-occupation n . For more details see [section 3](#). $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Labor Demand Effect of Monetary Policy - ONET, Wages to Wages, no lag

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.258 (0.407)	0.293 (0.404)	0.376 (0.401)		
Equilibrium Wages _{<i>i,c,t</i>}	0.934*** (0.003)	0.920*** (0.003)	0.922*** (0.003)	0.936*** (0.002)	0.937*** (0.002)
MP easing _{<i>t</i>} × Equilibrium Wages _{<i>i,c,t</i>}	-0.005 (0.039)	-0.008 (0.039)	-0.016 (0.039)	-0.002 (0.023)	-0.004 (0.023)
Obs.	6,916,841	6,916,841	6,912,065	6,916,143	6,911,267
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	226,653	226,653	226,599	226,649	226,593

This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP easing}_t \times X_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the wage posted by firm i , in commuting zone c in quarter t for the ONET-occupation n from the regional average salary. MP easing_{*t*} is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $X_{i,c,n,t-1}$ is defined as the log of the overtime average salary offered by firm i in commuting zone c at quarter $t-1$ for the ONET-occupation n . For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A6: Labor Demand Effect of Monetary Policy - ONET, Wages to Vacancies, no lag

	Log Vacancies _{<i>i,c,t</i>}				
	(1)	(2)	(3)	(4)	(5)
MP easing _{<i>t</i>}	0.180*** (0.035)	0.233*** (0.035)	0.233*** (0.034)		
Equilibrium Vacancies	-0.107*** (0.018)	-0.042*** (0.011)	-0.032** (0.013)	-0.001 (0.010)	0.029** (0.012)
MP easing _{<i>t</i>} × Equilibrium Vacancies	0.114 (0.104)	0.011 (0.092)	-0.013 (0.092)	0.034 (0.090)	0.025 (0.084)
Obs.	3,152,508	3,152,508	3,147,173	3,150,958	3,145,411
Firm FE	✓	✓	✓	✓	✓
CZ FE	✓	✓			
ONET FE		✓		✓	
ONET × CZ FE			✓		✓
CZ × Time FE				✓	✓
No. Firms	102,487	102,487	102,393	102,450	102,350

This table reports estimates of the following regression: $\text{Log Wages}_{i,c,t} = \alpha + \beta \text{MP easing}_t \times X_{i,c,n,t-1} + \theta Z_{i,c,n,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Wages}_{i,c,t}$ is defined as the log deviation of the wage posted by firm i , in commuting zone c in quarter t for the ONET-occupation n from the regional average salary. MP easing_{*t*} is the negative of the monetary policy shock by Jarociński and Karadi (2020), and thus a positive value reflects monetary policy easing. $X_{i,c,n,t-1}$ is defined as the cumulative vacancy share of firm i in commuting zone c at quarter $t-1$ for ONET-occupation n . For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** p<0.01, ** p<0.05, * p<0.1.

Table A7: Robustness to the Choice of Monetary Policy Shock

	Log Vacancies _{<i>i,c,t</i>}					
	(1)	(2)	(3)	(4)	(5)	(6)
MP easing _{<i>t</i>} × Equilibrium Vacancies _{<i>i,c,t</i>}	6.014*** (1.437)	2.525*** (0.625)	-0.065*** (0.015)	0.059*** (0.013)	0.039*** (0.010)	0.082*** (0.020)
R-squared	0.368	0.368	0.368	0.368	0.368	0.368
Obs.	15,791,442	15,791,442	15,069,930	15,069,930	15,069,930	15,069,930
Firm FE	✓	✓	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓	✓	✓
Shock	Nakamura and Steinsson (2018)	Bu et al. (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP easing}_t \times \text{LMP}_{i,c,t-1} + \theta X_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP easing_t are different monetary policy shocks, including Nakamura and Steinsson (2018), Bu et al. (2021) and Jarocinski (2021), in which a positive value reflects monetary policy easing. The four shocks from Jarocinski (2021) are: a standard contractionary monetary policy shock (column 3), a forward guidance shock that reflects a commitment future policy rates (column 4), a shock to longer-term treasury yields mostly affected by asset purchase announcements (column 5), and a forward guidance shock that captures the stance of future monetary policy in the sense of a prediction of the appropriate stance of policy (column 6). $\text{LMP}_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in the corresponding commuting zone c at quarter $t - 1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Robustness to the Choice of Monetary Policy Shock

	Log Vacancies _{<i>i,c,t</i>}					
	(1)	(2)	(3)	(4)	(5)	(6)
MP easing _{<i>t</i>} × Equilibrium Wages _{<i>i,c,t</i>}	-0.025*** (0.005)	-0.005** (0.002)	-0.0001 (0.0001)	-0.00002 (0.00005)	-0.0002** (0.00007)	0.0001 (0.0001)
R-squared	0.445	0.445	0.443	0.443	0.443	0.443
Obs.	4,454,151	4,454,151	4,135,037	4,135,037	4,135,037	4,135,037
Firm FE	✓	✓	✓	✓	✓	✓
CZ × Time FE	✓	✓	✓	✓	✓	✓
Shock	Nakamura and Steinsson (2018)	Bu et al. (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)	Jarocinski (2021)

This table reports estimates of the following regression: $\text{Log Vacancies}_{i,c,t} = \alpha + \beta \text{MP easing}_t \times \text{LMP}_{i,c,t-1} + \theta X_{i,c,t} + \gamma_{i,t} + \gamma_{c,t} + \varepsilon_{i,c,t}$, where $\text{Log Vacancies}_{i,c,t}$ is defined as the log number of vacancies posted by firm i , in commuting zone c in quarter t . MP easing_t are different monetary policy shocks, including Nakamura and Steinsson (2018), Bu et al. (2021) and Jarocinski (2021), in which a positive value reflects monetary policy easing. The four shocks from Jarocinski (2021) are: a standard contractionary monetary policy shock (column 3), a forward guidance shock that reflects a commitment future policy rates (column 4), a shock to longer-term treasury yields mostly affected by asset purchase announcements (column 5), and a forward guidance shock that captures the stance of future monetary policy in the sense of a prediction of the appropriate stance of policy (column 6). $\text{LMP}_{i,c,t-1}$ is defined as the cumulative vacancy share of firm i in the corresponding commuting zone c at quarter $t - 1$. For more details see section 3. $\gamma_{i,t}$ are firm-time, $\gamma_{c,t}$ are commuting-zone(CZ)-time fixed effects. Standard errors are double clustered at the firm and commuting zone level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Appendix Proofs and Propositions

B.1 Proposition 1

*Firms with greater classical monopsony power (larger γ_i) post **less** vacancies and pay lower wages in equilibrium.*

To prove Proposition 1, we consider the derivative of equilibrium vacancies and wages to γ :

$$(w_{it})'_\gamma = -\frac{1 - \alpha}{(1 + \gamma)^2} m$$

$$(v_{it})'_\gamma = -\frac{2}{1 + \gamma} - \log\left(\frac{n}{N}\right)$$

The first equation is unambiguously negative, indicating that firms with less substitutable labor markets (larger γ_i) pay more marked-down wages as discussed in the text.

The second equation is negative if the following condition is fulfilled (adding back subscripts for firm level outcomes):

$$\frac{2}{1 + \gamma_i} + \frac{1}{\gamma_i} \left(\alpha_i \log \left(\frac{v_i}{n_i} \right) + \log \left(\frac{w_i}{W} \right) - \log \left(\frac{L}{N} \right) + \log \left(\frac{D^{-1}}{1 + \gamma_i} \right) \right) > 0 \quad (16)$$

$$\text{where } \frac{v_i}{n_i} = \left(\alpha \frac{m_i}{c} \right)^{\frac{1}{1-\alpha_i}} \quad (17)$$

$$\alpha_i = 1 - \nu_i(1 + \gamma_i) \quad (18)$$

To evaluate equation (18), we first take aggregate values for $\frac{L}{N}$ from FRED.²⁰ We then bound firm-level labor market tightness, $\frac{v_i}{n_i}$, by imposing a lower bound of 2.15 on the marginal revenue from an additional worker relative to the cost of posting a vacancy, $\frac{m_i}{c}$, meaning that the marginal revenue from labor is approximately double the cost of posting a vacancy, which we believe is quite conservative.²¹ Then we assume that w/W equals the ratio of the firm-level markdown, ν_i and aggregate markdown, D^{-1} .²² That leaves three objects to evaluate: the markdown, ν_i , a dispersion term which is a measure of weighted average γ_i across firms, D^{-1} , and γ_i itself. These three objects can be mapped to functions of α_i and γ_i . However, direct estimates for firm-level γ_i or α_i are not available from the literature, meaning we cannot obtain bounds for either of them separately.

Therefore and given the above, our approach relies on finding minimum values of γ_i such that equation (18) holds, taking ν_i from a range of estimates from recent papers, [Berger et al. \(2022\)](#) and [Yeh et al. \(2022\)](#),²³ and making conservative assumptions about the value of D^{-1} .²⁴

Equation (18) holds if: (i) γ_i is larger than 0.193 (elasticity of substitution below 5.18) using estimates for the labor supply elasticity, and hence the markdown, from [Berger et al. \(2022\)](#), or (ii) γ_i is larger than 0.4299 (elasticity of substitution below 2.33) using the estimated markdown range from [Yeh et al. \(2022\)](#).

These bounds seem to be well within the acceptable range suggested by the limited estimates

²⁰The employment rate (relative to the labor force) $\left(\frac{L}{N} \right) = 0.943$.

²¹A lower bound on $\frac{m_i}{c}$ of 2.15 rationalizes the average labor market tightness, $\frac{V}{N}$ of 0.62 calculated from data on FRED for the US economy for the aggregate $\alpha = 0.34$ from [Barnichon and Figura \(2015\)](#), which is above the upper bound of values found by [Lange and Papageorgiou \(2020\)](#).

²²Here we are making an assumption that marginal revenue from worker does not correlate with labor market power parameters α and γ . In case when there is a negative correlation between markdown and marginal revenue from labor (more productive firms are larger and have more labor market power) our assumption delivers a lower bound. We take the average markdown for the US economy from [Hershbein et al. \(2022\)](#) at 0.65.

²³Bounds for ν_i are calculated from the estimates of the labor supply elasticity in Table 2 in [Berger et al. \(2022\)](#), using both contemporaneous and one-period-ahead results to get the largest possible range, and Table 1 in [Yeh et al. \(2022\)](#).

²⁴The dispersion term is such that $D^{-1} = \int_i (1 + \gamma_i) \left(\frac{n_i}{N} \right)^{\gamma_i} di$, with the right-hand side being the search-share weighted average of $(1 + \gamma_i)$. We take D^{-1} to be equal to the lowest of the γ_i we consider, which is conservative as it represents a lower bound for the possible values of D^{-1} .

available in the literature. For example, in [Marinescu and Wolthoff \(2020\)](#)²⁵, the estimated elasticity of search efforts to wage is around 0.78. Applying that to the markdown bounds from [Berger et al. \(2022\)](#) or [Yeh et al. \(2022\)](#) implies the lower bound for $\gamma_i = 1.48$ and $\gamma_i = 1.10$, respectively. Other related estimates, for instance, [Marinescu and Skandalis \(2021\)](#) estimate of the elasticity of labor participation to unemployment benefits suggests a substitution rate below 0.51, implying a lower bound for $\gamma = 1.96$. Note that the calculation of the bounds above assumed a weighted average γ_i as represented by D^{-1} that is much lower. Considering these larger values for γ_i mean that the condition would be even more likely to hold. Note additionally that the limiting factor for the lowest γ_i calculation is the largest α_i for firms with the low γ_i . If, for instance, we believe that firms are unlikely to have low γ_i if they have a large α_i , the bounds are once again, going to be relaxed. For instance, if the largest possible α_i for low γ_i firm is instead 0.2 – approximately at the average of the range suggested by [Lange and Papageorgiou \(2020\)](#) and delivered by some specifications in [Barnichon and Figura \(2015\)](#), the corresponding γ_i bounds are $\gamma_i > 0$ for [Berger et al. \(2022\)](#).

Our conclusion is that equation (18) is very likely to hold for *any* firm, which is not even necessary for Test 1 which requires the condition to hold for *most* firms.

B.2 Proposition 2

*Firms with greater search monopsony (larger α_i) pay lower wages, but post **more** vacancies in equilibrium.*

To prove Proposition 2, we consider the derivative of equilibrium vacancies and wages to α :

$$\begin{aligned}(w_{it})'_\alpha &= -\frac{1}{1+\gamma}m \\ (v_{it})'_\alpha \frac{\alpha}{v} &= \frac{\alpha}{(1-\alpha)^2} \log\left(\frac{\alpha m}{c}\right) \left(1 + \frac{1}{\gamma}\right) + \frac{1}{1-\alpha}\end{aligned}$$

The first equation is unambiguously negative, indicating that firms with larger matching elasticity pay more marked down wages as discussed in the text.

The second equation is positive if the following condition is fulfilled (adding back subscripts for firm level outcomes):

$$\alpha_i \left(1 + \frac{1}{\gamma_i}\right) \log\left(\frac{v_i}{n_i}\right) + 1 > 0 \quad (19)$$

$$\text{where } \frac{v_i}{n_i} = \left(\alpha_i \frac{m_i}{c}\right)^{\frac{1}{1-\alpha_i}} \quad (20)$$

$$\alpha_i = 1 - \nu_i(1 + \gamma_i) \quad (21)$$

As done to prove Proposition 1, we repeat the same procedure to obtain the lower bound for $\frac{v_i}{n_i}$. Then we use the range of estimates for the markdown from [Berger et al. \(2022\)](#) and [Yeh et al.](#)

²⁵[Marinescu and Wolthoff \(2020\)](#) explores an elasticity of the search intensity to the posted wage for online vacancies – making it very close to the case that we consider in this paper.

(2022) to judge whether equation (21) holds. The condition holds if γ is larger than 0.269 for both Berger et al. (2022) and Yeh et al. (2022) (implying the elasticity of substitution of 3.72).

Recalling the conclusion from the proof of Proposition 1, γ needs to be larger than 0.269 using estimates from Berger et al. (2022) and larger than 0.4299 using estimates from Yeh et al. (2022). These acceptable ranges go far beyond the limited estimates that are suggested in the literature, with, for instance, Marinescu and Wolthoff (2020), implying γ at least 1.10.

B.3 Proposition 3

In a model without search monopsony and following a monetary policy shock, if firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, if firms have varying degrees of search monopsony power and don't change their wages differentially, vacancies of firms with greater search monopsony power react more.

Take equation (4):

$$\hat{v}_i = \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i} di} \right) + \hat{N}$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$.

If wages respond homogeneously as assumed in the proposition, $\hat{w}_i = \hat{w}$, then we can take them out of the integral in the numerator of the term on the right, which then is simply $-\hat{w}$ and can be canceled out with $+\hat{w}$ that just precedes it. Equation (4) then simplifies to:

$$\hat{v}_i = \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i - \frac{\int \lambda_i \frac{1}{\gamma_i} \frac{\alpha_i}{1 - \alpha_i} \hat{m}_i di}{\int \lambda_i \frac{1}{\gamma_i} di} \right) + \hat{N}$$

First, consider the case when firms only differ in the extent of their classical monopsony power, or γ_i but $\alpha_i = \alpha$ and $\hat{m}_i = \hat{m}$. In this case, vacancies would respond homogeneously, just like wages.

The second part of Proposition 3 can be proven by considering firms with the same γ and $\hat{m}$ but different α_i . Assuming wages do not respond differentially, the response of vacancies would depend positively on α_i .

B.4 Extension 1

Extension 1: *In the absence of heterogeneity in search monopsony and of heterogeneity in productivity or any other non-monopsony factor that would generate heterogeneous responses of wages to monetary policy and assuming that the substitution between labor markets is homogeneous of degree one such that γ_i only depends on the own search share $\frac{n_i}{N}$, heterogeneity in classical monopsony alone cannot generate an heterogeneous response of vacancies.*

Examples of this type of substitution include Kimball and Atkeson-Burstein oligopoly. The latter is the framework used in [Berger et al. \(2022\)](#).

Under this extension, γ is allowed to vary and the response to monetary policy becomes:

$$\hat{v}_i = \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di} \right) + \hat{N} \quad (22)$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{W L}$ and $\gamma^* = \gamma \left(\frac{\eta}{\gamma + 1} + 1 \right)$ and $\eta_i = \frac{\partial \gamma_i}{\partial \frac{n_i}{N}} \frac{n_i}{N} \frac{1}{\gamma}$

Under this extension, the zero hypothesis for no homogeneity in α_i remains the same: if matching efficiency is heterogeneous, the response of vacancies is heterogeneous as well. If vacancies respond heterogeneously to monetary policy, so that firms with higher equilibrium vacancies and lower equilibrium wages respond more – there must be heterogeneity in α_i across firms.

Before considering the extension, we prove the following lemma. Note that, for simplicity this lemma is formulated from the perspective of a consumer problem, but, as we will see in the main proof, a similar structure would emerge in the context of the model that we consider here.

Lemma 1 For homogeneous of degree one substitution between markets and if "natural pricing assumption holds", the log-deviations of quantities and wages (quantities and prices) are such that:

$$\begin{aligned} \int \frac{p_i y_i}{PY} \hat{y}_i di &= \hat{Y} \\ \int \frac{p_i y_i}{PY} \hat{p}_i di &= \hat{P} \end{aligned}$$

From natural pricing assumption have:

$$PY = \int_i p_i y_i di$$

From homogeneous of degree one substitution, the aggregation equation is given by the following functional:

$$F\left(\frac{y_i}{Y}\right) = 1$$

Assume additionally that this functional is both Gateaux and Fréchet differentiable.

Conduct a simple cost minimization to derive first-order conditions (λ is a Lagrange multiplier):

$$\int p \psi_i di - \lambda dF\left(\frac{y_i}{Y}; \frac{\psi_i}{Y}\right) = 0 \quad \forall \psi(i) \in C^\infty$$

$$dF\left(\frac{y_i}{Y}; \frac{\psi_i}{Y}\right) = \frac{\int p \psi_i di}{\lambda}$$

Substitute $\psi_i = y_i$ and use the natural pricing assumption to obtain:

$$dF\left(\frac{y_i}{Y}; \frac{y_i}{Y}\right) = \frac{PY}{\lambda}$$

Now consider the aggregation equation and some deviation $\hat{y}_i = \frac{\psi_i}{y_i}$ and $\hat{Y} = \frac{\Psi}{Y}$ from a point where cost minimization holds. Substitute the first-order conditions from the cost-minimization problem.

$$dF\left(\frac{y_i}{Y}; \frac{\psi_i}{Y}\right) - dF\left(\frac{y_i}{Y}; \frac{y_i}{Y^2}\Psi\right) = 0$$

$$\begin{aligned} \frac{\int p\psi_i di}{\lambda} - \frac{PY}{\lambda} \frac{\Psi}{Y} &= 0 \\ \frac{PY}{\lambda} \int \frac{p_i y_i}{PY} \frac{\psi_i}{y_i} di - \frac{PY}{\lambda} \frac{\Psi}{Y} &= 0 \end{aligned}$$

$$\hat{Y} = \int \frac{p_i y_i}{PY} \hat{y}_i di$$

Now use the natural pricing and the result for quantities to derive an equation for prices:

$$\hat{P} + \hat{Y} = \int \frac{p_i y_i}{PY} (\hat{p}_i + \hat{y}_i) di$$

$$\hat{P} = \int \frac{p_i y_i}{PY} \hat{p}_i di$$

End of Lemma 1

Coming back to the problem at hand, consider the labor income maximization problem:

$$\begin{aligned} WL &= \max_{n_i} \int w_i \left(\frac{v_i}{n_i'}\right)^\alpha n_i di \\ F\left(\frac{n_i}{N}\right) &= 1 \end{aligned}$$

Where F is a homogeneous of degree one differentiable functional, representing the substitution between labor markets.

Lemma 1 can be applied in this problem after a substitution:

$$\begin{aligned} \bar{W}N &= \max_{n_i} \int \bar{w}_i n_i di \\ F\left(\frac{n_i}{N}\right) &= 1 \end{aligned}$$

where $\bar{W} = W \frac{L}{N}$ and $\bar{w}_i = w_i \left(\frac{v_i}{n_i'}\right)^\alpha$

Therefore, the deviations of the aggregate search efforts N can be expressed as:

$$N = \int \frac{\bar{w}_i n_i}{\bar{W} N} \hat{n}_i di = \int \frac{w_i l_i}{LW} \hat{n}_i di = \int \lambda_i \hat{n}_i di$$

As in the baseline model, firm's first-order conditions can be expressed as:

$$\begin{aligned} w_{it} &= \frac{1 - \alpha_i}{1 + \gamma_{it}} m_{it} \\ v_{it} &= \alpha_i \frac{l_{it}}{c} m_{it} \end{aligned}$$

where $\frac{1}{\gamma_{it}} = \frac{\partial n_{it}}{\partial \bar{w}_{it}} \frac{\bar{w}_{it}}{n_{it}}$

Now consider the response to monetary policy (change in the marginal revenue from labor m):

$$\hat{v}_i - \hat{n}_i = \frac{1}{1 - \alpha_i} \hat{m}$$

where $\eta_i = \frac{\partial \gamma_i}{\partial \frac{n_i}{N}} \frac{n_i}{N} \frac{1}{\gamma_i}$.

Note the difference with the standard case: $\eta \neq 0$, indicating non-zero superelasticity (η itself is not superelasticity, but can be expressed in terms of it).

Consider the labor supply equation and the result of Lemma 1 to obtain:

$$\hat{n} = \frac{1}{\gamma^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m} + w_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + w_i \right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di} \right) + \hat{N}$$

Substitute into the optimal solution for vacancy postings to obtain:

$$\hat{v}_i = \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di} \right) + \hat{N}$$

Note that if there is no heterogeneity in α_i , there is no homogeneity in response of either vacancies or wages.

B.5 Extension 2

Extension 2: *Under decreasing returns to scale or imperfect competition in the product market; if, following a monetary policy shock, firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, under the same conditions, vacancies of firms with greater search monopsony power react more.*

Decreasing returns to scale will result in $\hat{m}$ depending on the level of employment within the

firm. So the response of vacancies and wages under flexible wages will become:

$$\hat{w}_i = \hat{m}_i = \hat{m}p_i - \kappa_i l_i$$

Under Cobb-Douglas production function with decreasing returns to scale $y = z l^a$, $\kappa_i = (1 - a)$. Under imperfect competition in the product markets, $\kappa_i = \frac{1}{\xi^d}$ with ξ^d being elasticity of demand for products of the firm. Note that in both cases κ_i is between zero and one.

In both those cases, the response of vacancies to monetary policy becomes:

$$\hat{v}_i = \frac{1}{1 - (1 - \kappa_i)\alpha_i} \hat{m}_i + \frac{(1 - \kappa_i)(1 - \alpha_i)}{1 - (1 - \kappa_i)\alpha_i} \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di} \right) + \hat{N}$$

As in the proposition 3, if there is no heterogeneity in wage response and only heterogeneity in γ_i^* , there will be no heterogeneity in vacancy response.

B.6 Extension 3

Under wage rigidity, if, following a monetary policy shock, firms with varying degrees of classical monopsony power do not change their wages differentially, vacancies of these firms also do not change differentially. However, under the same conditions, vacancies of firms with greater search monopsony power react more.

Under wage rigidity, the formula for the response of vacancy postings does not change:

$$\hat{v}_i = \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di} \right) + \hat{N} \quad (23)$$

where λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$ and $\gamma^* = \gamma \left(\frac{\eta}{\gamma + 1} \right)$.

First order conditions and the response of wages change, however, if there is still no differential wage response to monetary policy, there will be no differential response of vacancies to monetary policy in the model with only classical monopsony power. One example for the particular assumptions about wage rigidity is considered below.

Assume Calvo wage setting, so that each period wages are only adjusted with probability θ .

In this model, the first-order conditions change and are listed below. For the wage the optimal reset wage at time t is given

$$w_{it}^* = \frac{\sum_{\tau \geq t} \theta^{\tau-t} l_{i\tau} \xi_{i\tau}^w m_{i\tau}}{\sum_{\tau \geq t} l_{i\tau} (\xi_{i\tau}^w + 1)}$$

$$v_{it} = \xi_{it}^v \frac{l_{it}}{c} m_{it}$$

Log-linearization of the optimal response to monetary policy shock is given by:

$$\begin{aligned}\hat{w}_{it} &= \theta(1 - \theta) \sum_{\tau \geq t} \theta^{\tau-t} \left[\hat{m}_{i\tau} + \frac{-\gamma_i \eta_i}{\gamma_i + 1} (\hat{n}_{i\tau} - \hat{N}_t) \right] + (1 - \theta) \hat{w}_{it-1} \\ \hat{v}_i &= \frac{1}{1 - \alpha_i} \hat{m}_i + \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i - \frac{\int \lambda_i \frac{1}{\gamma_i^*} \left(\frac{\alpha_i}{1 - \alpha_i} \hat{m}_i + \hat{w}_i \right) di}{\int \lambda_i \frac{1}{\gamma_i^*} di} \right) + \hat{N}\end{aligned}$$

where $\eta = \frac{\partial \gamma}{\partial \frac{n}{N}} \frac{n/N}{\gamma}$, λ_i is the equilibrium payroll share, $\lambda_i = \frac{w_i l_i}{WL}$ and $\gamma^* = \gamma \left(\frac{\eta}{\gamma+1} \right)$.

C Vacancy Postings and Employment

C.1 Compustat Dataset

We merge Lightcast and Compustat data for a large number of publicly traded firms to analyze the relationship between vacancy postings and employment growth. We aggregate vacancy postings to the firm level and fuzzy merge the Lightcast firm name to Compustat. First, we execute a standard string cleaning procedure, removing excessive spacing, acronyms related to a company's business structure acronyms, special characters, and other unnecessary characters from the name. Given the string differences between the names of companies in Lightcast and Compustat, we used a two-layered merging technique consisting of exact matching, and Jaro-Winkler string distance matching (Jaro, 1989; Winkler, 1990). For the Jaro-Winkler fuzzy matching, we set a string distance threshold of 0.11, which maximizes the number of matches and their quality jointly. We obtain 8,231 firm matches from 14,983 firms in Compustat in the years between 2008 to 2019, of which 3,217 are exact matches, and 5,014 match using the Jaro-Winkler fuzzy matching. The merged companies represent 75% of sales, and 73% of employment of all companies in Compustat.

We follow the cleaning procedure of Ottonello and Winberry (2020) for the Compustat data closely. First, we keep only corporations incorporated in the US. In particular, we drop firms in the financial and utilities sectors, firm-quarter observations with acquisitions larger than 5 percent of assets, firm-quarter observations where the investment rate is in the top and bottom 1 percent of the distribution, the investment spell is shorter than 40 quarters, firm-quarter observations where liquidity, debt, and sales are outliers, firms with less than 10 million US dollars in assets, and that are in the sample less than 5 years. Since the quarterly version of Compustat does not have employment information, we linearly interpolate the annual employment data to quarterly data.

C.2 Results

So far, we have established that vacancy postings are more responsive to monetary policy when firms have more labor market power. Ultimately, what matters for monetary policy is em-

ployment and not vacancy postings. Unfortunately, detailed granular employment data on the firm-region-level are not publicly available.

To understand the relationship between vacancy postings and employment, we estimate the following regression equation:

$$\Delta Employment_{i,t} = \alpha_i + \alpha_t + \beta_1 \text{Log Vacancies}_{i,t} + \epsilon_{i,t} \quad (24)$$

where $\Delta Employment_{i,t}$ is the log change in employment of firm i between year t and $t - 1$ in Compustat.²⁶ $\text{Log Vacancies}_{i,t}$ is the log number of vacancies posted by firm i in year t from Lightcast. $\text{Log Vacancies}_{i,t}$ is defined in the same way as the dependent variable in ??, which allows us under certain assumptions to translate the effect of monetary policy on vacancies to an effect on employment based on the elasticity estimated in Equation 24. Figure 3 shows the result of Equation 24 in a scatterplot. The relationship between the number of vacancies and the percent change in employment is positive and statistically significant. Economically, a doubling in the number of vacancies ($\text{Log Vacancies}_{i,t} = 1$) is associated with a 0.74 percentage point stronger employment growth.

?? had shown that after four quarters, a firm with high labor market power increased its vacancy postings by a factor of 2 in response to a 10 basis point accommodative monetary policy shock. A firm with medium labor market power instead did not increase its vacancy postings. Translating the vacancy postings into employment growth, we need to multiply the log number of vacancies created by the coefficient on the elasticity of employment growth to vacancies. Consequently, a firm with high labor market power has $(0.74 * 2 =) 1.48$ percentage points stronger employment growth in response to the accommodative monetary policy shock. According to our estimates, a firm without labor market power does not exhibit stronger employment growth.

This back-of-the-envelope calculation makes several assumptions. First, we only have employment data for listed firms. For the calculation to be accurate, the elasticity needs to be the same for firms that we merge with Compustat and the firms that we do not merge. Second, the elasticity of employment growth with respect to vacancy postings may vary between firms with and without labor market power. For instance, monopsonists may post more vacancies but do not increase their actual hiring in response to an accommodative monetary policy shock, as more employees leave when labor market becomes tighter in response to the shock. However, we do not find evidence in favor of a differential elasticity for firms with differential degree of labor market power, suggesting that higher vacancy postings of firms with labor market power also reflect more intense hiring and employment growth. Even for firm with a large degree of labor market power there is a strong relationship between vacancy postings and employment growth

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²⁶The contemporaneous specification reflects the fact that most vacancies are filled well before a year passes, for instance, the average time to fill a vacancy stayed at approximately one month in the time period we consider per Job Openings and Labor Turnover Survey (JOLTS).

²⁷The positive relationship is robust to using other specifications, such as log-log equations.

To further verify the results hold for employment rather than vacancies, we estimate the dynamic response of employment in Compustat.

We estimate the following regression:

$$\sum_{h=0}^H \Delta \text{Employment}_{i,t+h} = \alpha_H + \beta_H \text{MP easing}_t \times \text{LMP}_{i,t-1} + \theta_H X_{i,t} + \gamma_i + \gamma_{t+H} + \varepsilon_{i,t+H}$$

where $\Delta \text{Employment}_{i,t+h}$ is defined as the log difference between employment in time $t+h$ and $t-1$. MP easing_t is the negative of the monetary policy shock by [Jarociński and Karadi \(2020\)](#), in which a positive value reflects monetary policy easing. $\text{LMP}_{i,t-1}$ is defined as one if the firm is in the top 5% of labor market power in at least one commuting zone. Standard errors are clustered at the firm level. Shaded areas represent a 95% confidence interval.

?? plots β_h of this regression. In response to a 10 bp monetary easing shock firms with labor market power increase their employment by 1-2 percentage point more than firms without labor market power within the first year. The result is consistent with the above back-of-the-envelope calculation, under which the response is around 1.5%.